

BIOLOGY GRADE 11: TAXONOMY II

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.3: Taxonomy II
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Kingdom Plantae (general characteristics of organisms in Divisions Bryophyta, Pteridophyta and Spermatophyta; include sub-divisions and classes of Spermatophyta) – Kingdom Animalia (general characteristics of organisms in Chordata and Arthropoda up to class level; include economic importance of arthropods) – Dichotomous keys (principles of construction, identification of organisms, and practical applications) – Viruses (general characteristics, conditions for their growth, and economic importance) – Projects: preparing herbarium specimens and mounting animal specimens or their images using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) outline the characteristics of organisms in the divisions of Kingdom Plantae, including Bryophyta, Pteridophyta and Spermatophyta (sub-divisions and classes of Spermatophyta); b) classify organisms into phyla and classes of Kingdom Animalia; c) construct dichotomous keys to identify plants and animals; d) relate viruses to their economic importance in health, agriculture and biotechnology; e) appreciate the diversity of plants and animals in nature.
Core Competencies	– Communication and Collaboration: The learner speaks clearly to peers while discussing distinguishing characteristics of various specimens (Lessons 1–8), working in groups to agree on classification criteria, build keys and present findings. – Critical Thinking and Problem Solving: Learners decide which characteristics are useful for classification, solve identification problems and justify classification decisions (Lessons 1, 4, 6 and 8). – Creativity and Imagination: The learner experiments with varied ideas while preparing a herbarium of plant specimens, and designs classification charts, dichotomous keys and biodiversity guides (Lessons 3, 4 and 8).
Core Values	– Respect: The learner is considerate of others' opinions while brainstorming on the characteristics of viruses and their economic importance, and listens to alternative classification ideas. – Responsibility: The learner cares for the environment while collecting specimens of monocot and dicot plants, and handles specimens and learning resources carefully. – Excellence: Learners strive for accuracy when identifying and classifying organisms.
Science & Engineering	– Asking questions: Learners generate questions for the Driving Question Board about how organisms are grouped and why

Practices	viruses matter. – Developing and using models: Learners build and revise classification charts and dichotomous keys across the unit. – Analyzing and interpreting data: Learners compare observed features of plants and animals in tables to find patterns. – Constructing explanations: Learners use evidence from specimens to justify why an organism belongs in a division, phylum or class. – Engaging in argument from evidence: Learners defend classification decisions to peers and revise them when given new evidence. – Obtaining, evaluating and communicating information: Learners use textbooks, readings and diagrams to learn about viruses and present group keys.
Pertinent & Contemporary Issues (PCIs)	– Environmental Conservation: The learner conserves biodiversity by responsibly collecting organisms for the preparation of herbarium specimens. – Animal Welfare: The learner handles animals with care while observing and grouping them into Phylum Chordata and Arthropoda using distinguishing characteristics. – Health and disease awareness: Learners relate viruses to human, animal and plant diseases and their prevention. – Environmental sustainability and responsible use of biological resources: Learners appreciate why biodiversity should be documented and used wisely.
Career Connections	– Biology teacher: teaches classification and guides learners in identifying organisms. – Botanist: classifies plants into divisions and prepares herbarium collections, for example at the National Museums of Kenya. – Zoologist: groups animals into phyla and classes. – Ecologist: uses classification to study biodiversity in habitats such as the Rift Valley and Lake Victoria. – Taxonomist: names, describes and organises organisms using keys. – Agricultural scientist: identifies crop plants, pests and plant viruses affecting maize and beans. – Veterinary professional: identifies animals and viral diseases of livestock. – Wildlife officer: identifies species in parks and reserves such as Maasai Mara. – Environmental scientist: monitors biodiversity and sustainability. – Laboratory technologist: handles specimens and identifies disease-causing agents such as viruses. – Research scientist: studies viruses and biodiversity, including in biotechnology.
Focus for Lessons	This unit uses familiar organisms from the school and local environment (maize, beans, fern, moss, mushroom, mosquito, earthworm, chicken, goat) to build understanding of how plants, animals and viruses are classified. Learners first sort organisms using their own criteria, then compare these with scientific classification of Kingdom Plantae and Kingdom Animalia, practise using and constructing dichotomous keys, and explore viruses. Local specimens, pictures and textbooks are used so activities do not depend on internet access, and learners may discuss in Kiswahili or mother tongue before recording scientific terms in English.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we identify and classify living organisms accurately? KICD KEY INQUIRY QUESTIONS: 1. How are plants and animals classified? 2. How are viruses important in nature?
Anchoring Phenomenon	The teacher displays or takes learners outside to observe a collection of organisms found around the school and local environment: a maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat. All are living things, yet they differ greatly in structure, size, body covering, way of moving and reproduction. Learners are

	asked: "If all these organisms are living things, how can scientists organise them into groups so that any person anywhere can identify and study them?" It is puzzling because learners' own groupings (for example by size, colour, habitat or usefulness as food) differ from group to group, and no single grouping works for everyone. This shows the need for a standard system based on shared, observable characteristics.
Storyline Thread	Lesson 1: Why classify organisms? Organisms can be grouped using similarities and differences, but scientists require a standard system of classification. Learners observe local organisms/pictures, independently classify them, compare groupings, discuss differences and begin the DQB. Resources: local specimens, organism pictures, manila paper, markers, textbooks, video, exit ticket. Leads to: How do scientists decide which characteristics should be used for classification? Lesson 2: Kingdom Plantae — Bryophyta. Bryophytes have distinctive characteristics that separate them from other plant groups. Learners observe moss or pictures/specimens, examine structures, record similarities and differences, and make a labelled drawing. Resources: moss specimens/pictures, hand lenses, charts, textbooks, video. Leads to: What makes other plants different from bryophytes? Lesson 3: Pteridophyta and Spermatophyta. Pteridophytes and spermatophytes have features that distinguish them from bryophytes and from each other. Learners examine ferns, maize, beans or other available plants, compare roots, stems, leaves and reproductive structures, and construct a comparison table. Resources: ferns, maize/bean plants, specimens/pictures, charts, videos. Leads to: How can we classify plants systematically when we encounter an unknown plant? Lesson 4: Using dichotomous keys. A dichotomous key uses pairs of contrasting characteristics to identify organisms. Learners study a sample key, use it to identify local plants, and create a simple key for collected plant specimens. Resources: plant specimens, prepared keys, worksheets, rulers, video. Leads to: Can the same method be used to identify animals? Lesson 5: Kingdom Animalia — major phyla. Animals can be classified into major phyla according to characteristic body features. Learners observe pictures/specimens of animals, identify body characteristics, group organisms and justify placement in different phyla. Resources: animal specimens/pictures, charts, textbooks, video. Leads to: How are animals within the same phylum further classified? Lesson 6: Classes of Kingdom Animalia. Animal phyla can be divided further into classes using more specific characteristics. Learners examine examples of animals, complete classification tables, compare body covering, limbs, respiration and reproduction, and defend classifications. Resources: animal pictures/specimens, charts, classification cards, video. Leads to: How can we identify an unknown animal accurately? Lesson 7: Dichotomous keys for animals and viruses. Dichotomous keys can identify organisms using observable contrasting features; viruses have distinctive characteristics and importance. Learners construct and use an animal dichotomous key, discuss virus characteristics and their effects, and revise initial ideas. Resources: animal cards, prepared keys, virus diagrams, textbook/readings, video. Leads to: How can our classification system help us explain biodiversity and the importance of viruses? Lesson 8: Synthesis and explanation. Classification provides an organised system for identifying, studying and communicating about biodiversity. Learners revisit the DQB, complete a final classification challenge, present group models/keys and use evidence to answer the driving question. Resources: organism cards/specimens, charts, learners' notes, rubric, video. Leads to: How can classification knowledge be applied outside the classroom?
Supporting Phenomena	– Two groups of learners sort the same set of organisms differently (by habitat versus by body structure), and neither can easily tell another group how to identify an unknown specimen. – A fern and a moss grow side by side on a damp wall or shaded stream bank, yet one has true roots, vascular tissue and fronds while the other is small and lacks these structures. – Maize and bean plants grow together in a Kenyan farm intercrop, but their leaves, roots and seeds differ (monocot versus dicot). – A farmer in Kisumu or Nairobi notices that maize leaves show streaking caused by a virus, while a mosquito and a housefly carry different diseases: viruses are not like other organisms and cannot be treated as the rest.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute lessons)): Why classify organisms?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with a collection of local organisms that they must group themselves. Learners discover that different groups use different criteria, and that a standard system based on shared, observable characteristics is needed so anyone can identify and study organisms.
Knowledge	Organisms can be grouped using similarities and differences. Groupings based on size, colour, habitat or usefulness vary from person to person. Scientists therefore need a standard system of classification based on shared, observable characteristics. This links to KICD outcome (c): classify organisms into phyla and classes of Kingdom Animalia and Kingdom Plantae, and prepares for outcomes (a) and (b) on Divisions of Kingdom Plantae.
Skills	Observing and recording structural features of organisms; classifying using self-chosen criteria; communicating reasoning (SEP: Constructing Explanations); comparing and critiquing groupings (SEP: Engaging in Argument from Evidence); asking questions (SEP: Asking Questions); developing an initial model (SEP: Developing and Using Models). Core competency: Communication and Collaboration - the learner speaks clearly to peers while discussing distinguishing characteristics of various specimens.
Attitudes	Respect: the learner is considerate of others' opinions when groupings differ. Responsibility: the learner cares for the environment while collecting and handling specimens. Appreciation of the diversity of plants and animals in nature (KICD outcome e). PCIs: Environmental Conservation - collect only common, safe organisms responsibly; Animal Welfare - handle animals with care and return them to where they were found.
Key Inquiry Question	How can we identify and classify living organisms accurately? (KICD inquiry: How are plants and animals classified?)
Purpose in Storyline	Lesson 1 creates the need for the unit. Learners see that all the organisms are living things, yet their own groupings conflict. This raises the question for Lesson 2: How do scientists decide which characteristics should be used for classification? The initial model and the DQB are revisited in Lessons 4 and 8.
Safety Notes	Do NOT collect unknown, venomous or dangerous organisms. Do not touch wild mushrooms with bare hands, and never taste any specimen. The teacher should handle the mushroom, lizard and mosquito (use a pictured or dead specimen). Wash hands after handling soil, earthworms, snails or plants. Handle live animals (earthworm, snail, lizard) gently and return them to their habitat after the lesson. Use photographs for any organism that is unavailable or unsafe. Keep learners supervised outdoors and away from roads, water edges and thorny areas.

B. LESSON OVERVIEW

Learners walk around the school compound (or view a table display) to observe a maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat (using photographs where specimens are unavailable). They first predict how scientists would group them, then observe closely and independently sort the organisms using their own criteria. Groups put their groupings on manila paper and do a gallery walk to compare. They notice that groupings by size, colour, habitat or food value differ, and that no single grouping works for everyone. Through Think-Pair-Share, learners explain why a standard system is needed. The class opens the Driving Question Board (DQB) with their questions and draws an initial model of how organisms could be grouped. An exit ticket checks understanding and leads to Lesson 2: How do scientists decide which characteristics should be used for classification?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 23 The Classification Of Living Things Source: Seavuria  Search ARES for similar videos  EXERCISE: Classification Of Organisms Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners stand around the display of organisms or photographs: maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat. The teacher asks: 'If all these organisms are living things, how can scientists organise them into groups so that any person anywhere - in Kisumu, Mombasa or Nairobi - can identify and study them?' Individually, learners write a prediction in their notebooks: (1) how many groups they think are needed, (2) what they would use to decide the groups, and (3) whether their groups would match their neighbour's. They discuss ideas first in Kiswahili or mother tongue if helpful, but record key words in English (e.g., 'plant', 'animal', 'group'). They then Think-Pair-Share their predictions with a partner.	Before class, gather safe, common local specimens or pictures and arrange them on desks or a table with number tags 1-13. Say: 'All of these are living things. Look carefully for 2 minutes without talking. What do you notice that is the same? What is different?' Pose the phenomenon question aloud and write it on the board. Give 1 minute silent writing time, then say: 'Turn to your partner and share your prediction.' Give WAIT TIME of 10-15 seconds after asking 'What would you use to make your groups?' before taking any answers. Cold-call 4 learners (different parts of the room) to share a partner's idea, not their own: 'Tell us what your partner predicted.' Record criteria on the board without judging them as right or wrong (e.g., size, colour, habitat, food value). Do NOT give the term 'taxonomy' yet.	Predict-Observe-Explain (POE) - Predict step. Recording predictions before evidence makes learners' prior ideas visible and gives them something to compare with later evidence, which promotes conceptual change. Think-Pair-Share lets every learner voice an idea before whole-class discussion.	Check notebooks as you move around: Do 80% of learners write a prediction that includes at least one criterion for grouping? Listen for everyday criteria (size, colour, 'where it lives', 'what we eat') and note which learners mention structure or body features. Note learners who think grouping is obvious and unambiguous, to challenge in the Explain phase. Tally criteria on the board to use later.
Observe Phase  VIDEO: 23 The Classification Of Living Things Source: Seavuria  Search ARES for similar videos  HTML: Animal Classification (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES	In groups of 4-5, learners observe the organisms closely (outdoors in the school compound or at the display) for about 15 minutes. Each learner first works independently to draw or list the 13 organisms and record at least two observable features for each (e.g., has leaves, has legs, has wings, body covering, way of moving, how it gets food, how it reproduces if known). Then each learner independently groups the organisms using his or her own criteria and writes why each organism was placed in a group. After that, the group compares individual groupings and agrees on	Distribute observation sheets and manila paper and markers. Set safety expectations: 'We observe with our eyes first. We do not touch the mushroom, and we handle the earthworm, snail and lizard gently - they are living things like us.' Circulate and ask probing questions, such as: 'Why did you put the mushroom with the plants? What made you decide that?' and 'Where would you put the moss - with the grass or somewhere else? Why?' Use WAIT TIME of 10-15 seconds after each question. Prompt groups that finish early: 'Can you make a different grouping using a different rule?' Do NOT tell	Predict-Observe-Explain (POE) - Observe step. Learners gather first-hand evidence from local organisms. Independent classification before group work ensures every learner builds their own ideas, and the clash between individual and group groupings begins to show that criteria matter.	Observable indicators: (1) each learner's notebook lists at least 2 features per organism; (2) each learner has an individual grouping with a stated reason; (3) groups' manila papers show clear labelled groups with reasons. Look for misconceptions such as 'a mushroom is a plant because it grows in soil', 'the snail is an insect', or 'a mosquito and a butterfly are different because one is harmful'. Record these on a checklist for follow-up.

for similar readings	one grouping, which they draw on manila paper with a title for each group and a reason for it. Groups handle animals gently and return them to their habitat.	them the scientific groups. Make sure each group sticks its manila paper on the wall at the end. Cold-call 3 groups to state the rule they used in one sentence. Use textbook illustrations or pictures for unavailable organisms, and let groups share textbooks.		
Explain Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Animal Classification Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners do a gallery walk, visiting at least 3 other groups' manila paper. At each, they note one similarity and one difference compared with their own grouping on sticky notes or in their notebooks. Back in their groups, they discuss: Why did groups differ? Which rule is best? Could someone in another county use our grouping and get the same result? Learners then Think-Pair-Share their conclusion: 'Why do scientists need a standard system of classification?' They give examples, e.g., a goat and a chicken would be grouped together as 'food' or 'farm animals' by some groups but separated by those using legs, body covering or way of reproducing. They watch a short video clip (if available) or read a textbook section on why organisms are classified, and add to their explanations.	Say: 'Walk quietly to another group's work. Find one grouping that is the same as yours and one that is different.' After the gallery walk, ask: 'What did you notice? Which organisms were hardest to place?' Give WAIT TIME of 10-15 seconds. Press for reasoning: 'Is grouping by size always reliable? What about the caterpillar that becomes a butterfly?' and 'If a visitor from Kericho used your grouping, would they place the mushroom exactly where you did?' Cold-call 5 learners to share an idea from their pair. Highlight that groupings by colour, size, habitat or usefulness vary and none works for everyone. Then lead the class to state: 'Organisms can be grouped using similarities and differences, but scientists require a standard system of classification based on shared, observable characteristics.' Introduce the term 'classification' and write it on the board. If a video is available, play it for 3-4 minutes; otherwise, ask groups to read the textbook introduction on classification (shared textbooks) and summarise it in 2 sentences. Do not introduce Kingdom names in detail yet.	Think-Pair-Share and Gallery Walk - learners compare evidence across groups and construct a shared explanation. Seeing conflicting groupings creates cognitive conflict that motivates the need for a standard system (SEP: Constructing Explanations; Engaging in Argument from Evidence). This completes the POE cycle by comparing predictions with observations.	Listen for learners who say: 'The groupings differ because we used different rules' and 'We need the same rules for everyone.' Ask 5 random learners to state in one sentence why a standard system is needed. Indicator of success: at least 75% can state that a standard system uses shared, observable characteristics so anyone anywhere gets the same result. Note any who still say one grouping is simply 'correct' without justification.
Driving Question Board (DQB) Creation  VIDEO:	Learners return to their groups and write at least two questions each on sticky notes or strips of paper about classifying organisms. Examples: 'How do scientists	Display a large chart titled 'DQB: How can we identify and classify living organisms accurately?' Say: 'Write questions you still have. Questions starting with How or	Driving Question Board - learners' own questions give purpose and direction to the unit, make learning visible and allow them to track growing understanding. Sorting	Check that each learner has contributed at least 1 question. Look at the quality of questions: Are they about features, groups or the process of classification

24 Six Kingdoms Of Classification Related topic: not an exact match for this lesson. Source: Seavuria Search ARES for similar videos  HTML: Plant Classification (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	decide which features to use?', 'Is a mushroom a plant or an animal?', 'Why do moss and fern look so different from maize?', 'What groups do the mosquito and the butterfly belong to?', 'Are viruses living?' Groups post questions on the DQB under the driving question: 'How can we identify and classify living organisms accurately?' The class sorts questions into categories (e.g., characteristics used, plant groups, animal groups, tools for identification, viruses). Learners also connect the KICD inquiry questions: 'How are plants and animals classified?' and 'How are viruses important in nature?'	Why are very useful.' Give WAIT TIME of 10-15 seconds, then ask: 'What are you still wondering about these organisms?' Cold-call 4 learners to read one question and place it on the board. Cluster similar questions with the class, asking: 'Which questions go together? What name could we give this group?' Point out the question: 'How do scientists decide which characteristics should be used for classification?' and say we will begin there in Lesson 2. Add the KICD inquiry questions if the class has not raised them. Keep the DQB visible and accessible all unit.	questions shows patterns and sequences the next lessons.	(higher quality) versus recall of names only? Observable indicator: at least 3 question clusters emerge and one links to choosing characteristics. Note respectful listening to others' questions (Respect value).
Model Building Phase  VIDEO: 23 The Classification Of Living Things Source: Seavuria Search ARES for similar videos  EXERCISE: Plant Classification Practice Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	Individually, learners draw an initial model on a half-page of paper or in their notebooks. It shows (1) the 13 organisms, (2) their own best grouping, (3) the rule(s) they used, and (4) a note on where their model might fail for someone else (e.g., 'my grouping by habitat puts the lizard and the earthworm together'). They add a short label such as 'What I still don't know'. They then complete the exit ticket: (a) Write one reason why scientists need a standard system of classification. (b) Give one observable characteristic that could be used to group the maize, bean plant and grass together. (c) Write one question you still have. Models are kept in a folder for revision in Lessons 4 and 8.	Say: 'Draw your best current model of how these organisms can be grouped. You will change it later as you learn more - so it is fine if it is not perfect.' Give WAIT TIME of 10-15 seconds after explaining and invite questions. Circulate and ask: 'What rule are you using? Where does your model not work?' Cold-call 3 learners to share the 'weak point' in their model. Hand out the exit ticket and collect it at the door. Close by naming the next question: 'How do scientists decide which characteristics should be used for classification?' Collect models and store them safely for later revision. Ask learners to carefully return any specimens to where they were found and wash hands.	Model revision (initial model) - learners record their current thinking as a model that includes its limitations. Revisiting it in Lessons 4 and 8 helps them see how their understanding grows with evidence. Exit ticket gives each learner a private way to show understanding.	Review models and exit tickets using this checklist: (1) model shows a grouping with a stated rule; (2) learner identifies at least one limitation; (3) exit ticket gives a reason for a standard system (e.g., 'so everyone groups the same way'); (4) names an observable characteristic (e.g., 'have leaves and roots'); (5) asks a question. Success criterion: 75% meet at least 3 of 5. Sort tickets into 'ready', 'needs a nudge' and 'needs reteaching' groups to plan Lesson 2.

D. TEACHER REFLECTION

After the lesson, reflect: Did learners truly attempt grouping independently before comparing? Which criteria appeared most (size, colour, habitat, usefulness, structure)?

Which organisms caused disagreement (mushroom, moss, snail, mosquito)? Did the specimens or pictures work well and were they safe? Did learners using Kiswahili or mother tongue participate more? What misconceptions need to be addressed in Lesson 2 (e.g., mushroom as a plant, snail as an insect)? Did every learner contribute a question to the DQB? How will I use the exit ticket results to adjust the start of Lesson 2?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed 13 local organisms (maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) and noted that they differ in structure, size, body covering, movement and reproduction. Different groups sorted them in different ways (by size, colour, habitat, usefulness).
What did I learn?	Organisms can be grouped using similarities and differences, but groupings made with different rules do not agree. Scientists need a standard system of classification based on shared, observable characteristics.
How does this explain the phenomenon?	Because everyone chose different rules, no single grouping worked for everyone. A standard system uses the same observable characteristics so that any person anywhere can identify and study an organism and get the same result. We still need to find out how scientists decide which characteristics to use.

LESSON 2 (80 minutes (2 x 40-minute periods)): Kingdom Plantae — Bryophyta

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence that bryophytes (mosses and liverworts) have distinctive shared characteristics that separate them from other plant groups, and use that evidence to improve their classification of the school-environment organisms (moss, fern, maize, bean, grass) and advance the driving question: How can we identify and classify living organisms accurately?
Knowledge	By the end of the lesson the learner should be able to outline the general characteristics of organisms in Division Bryophyta: small size, no true roots, stems or leaves, rhizoids for anchorage, no vascular tissue (xylem and phloem), need for moist habitats, spore-bearing capsules on stalks, and dependence of the sporophyte on the gametophyte.
Skills	Observing with a hand lens; recording similarities and differences in a table; making an accurate labelled biological drawing with title, scale/magnification, and straight label lines; communicating findings clearly to peers (SEP: Analysing and Interpreting Data; Developing and Using Models).
Attitudes	Appreciate the diversity of plants in nature; Respect: be considerate of others' opinions during discussion; Responsibility: care for the environment when collecting specimens; Environmental Conservation: collect only a small amount of moss and return the rest of the habitat undisturbed. Core competencies: Communication and Collaboration - the learner speaks clearly to peers while discussing distinguishing characteristics of various specimens.
Key Inquiry Question	How are plants and animals classified?
Purpose in Storyline	Lesson 1 showed that learners' own groupings (by size, colour, habitat, use as food) conflict and that a standard system based on shared observable characteristics is needed. Lesson 2 gives the first piece of such evidence: one plant group, the bryophytes, can be defined by a set of observable features. This creates the need for Lesson 3: What makes other plants different from bryophytes?
Safety Notes	Collect moss only from safe, accessible damp walls, tree bases, or shaded ground within the school compound; do not climb or go near drains, deep water, or snake-prone long grass. Take only a small patch and replace the rest. Wash hands after handling specimens and soil. Do not taste any plant or mushroom. Hand lenses must never be used to look at the sun. If a learner has allergies to moulds or spores, they should record data rather than handle specimens.

B. LESSON OVERVIEW

Learners return to the school-environment collection from Lesson 1 (maize, bean, grass, fern, moss, mushroom and others) and focus on the moss, which many groups placed with 'grass' or 'small green things'. They predict how moss differs from maize or bean plants, then examine real moss (or pictures and charts where unavailable) with hand lenses, recording similarities and differences in a table and producing a labelled drawing. A gallery walk lets groups compare drawings and findings, and Think-Pair-Share helps them reason about why moss grows only in damp places and stays small. Learners add new questions to the Driving Question Board and revise their group classification model to include a 'bryophyte' category defined by observable features. The lesson ends by asking: What makes other plants different from bryophytes?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy bryophyta moss liverwort  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: taxonomy bryophyta moss liverwort	Learners sit in their Lesson 1 groups of 4-5 with their earlier classification charts of the school organisms (maize, bean, grass, fern, moss, mushroom, etc.). A moss sample, or picture, is shown. On a sheet, each learner privately writes a prediction: 'Is moss more like maize, grass, or fern? What parts does it have (roots, stem, leaves, flowers)? Why does it grow where it does?' They then do a Think-Pair-Share, using Kiswahili or mother tongue if helpful, but write key terms (root, stem, leaf, spore) in English.	Hold up the moss and say: 'In Lesson 1 your groups placed this moss in different places. Today we ask: is moss just a very small grass, or is it something different?' Give 1 minute silent writing time. Say: 'Write your prediction and one reason. I will not tell you if you are right yet.' Then say 'Turn to your partner and share.' Allow 2 minutes. WAIT TIME 10-15 seconds after asking 'Where on the school compound did we find moss and what does that tell us?' before taking responses. Cold-call 3 learners from different groups using the roll book, not volunteers, and record their predictions on the board under 'Our predictions' without judging them. Ask: 'Who has a different idea?'	Predict-Observe-Explain (POE) and Think-Pair-Share: Making predictions before observation reveals learners' existing ideas (e.g. that moss has roots and flowers like maize) and creates a reason to look carefully. Pair sharing lets every learner rehearse ideas before whole-class talk, so they are invested in checking their prediction against evidence.	Check prediction sheets while circulating. Look for: (1) a prediction that names at least two plant parts; (2) a reason linked to an earlier observation; (3) any misconception such as 'moss has roots and flowers' or 'moss is a fungus like mushroom'. Tally on the board how many predict moss is 'like grass', 'like fern', or 'different from both' to revisit after observation.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy bryophyta moss liverwort  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: taxonomy bryophyta moss liverwort	Learners collect a small, teacher-approved patch of moss from a damp shaded area near the school (or receive prepared specimens, pictures, charts and textbook illustrations where moss is unavailable; a short video may be used if available but is not required). Using hand lenses, they examine: the overall size and colour; the leaf-like structures (are they thin, one cell thick, with veins?); the stem-like structure; the thread-like rhizoids at the base and whether there are real roots; and, where present, the stalk and spore capsule. They compare with a maize or bean seedling (and a fern frond if available) to record similarities and differences in a table with columns: Feature Moss Maize/Bean Fern. They measure heights with a ruler, note	Before going out, state the safety and conservation rules: 'Take only a small patch the size of your palm, put the rest back, and stay within the marked area.' Give each group a hand lens, a ruler, a plastic bag or tray, and a data table. Circulate and ask: 'What do you see at the base of the moss? Is that the same as the maize root? How do you know?' and 'What do you notice about the thickness of the leaves?' Use WAIT TIME of 10-15 seconds after each question. Prompt for evidence: 'Show me on the specimen.' Model drawing skills briefly: 'Use a sharp pencil, single clear lines, no shading, label lines must not cross.' If groups lack specimens, give each a picture or chart and ask them to point out features using the textbook. Allow	Structured observation with data table and scientific drawing: Comparing moss side by side with maize, bean and fern makes learners focus on shared, observable features instead of size or colour alone. Recording evidence in a table converts observations into data that can be used for classification, building the idea that groups are defined by sets of characteristics.	Observe whether learners: (1) use hand lenses correctly and refer to specific structures; (2) record at least four differences between moss and maize/bean in the table (e.g. rhizoids vs true roots, no flowers, small size, no vascular tissue visible, spore capsule vs seeds); (3) produce a drawing with title, magnification, accurate proportions and neat label lines. Use a 3-point checklist (Meets / Partly / Not yet) and note groups needing support.

	the habitat (damp, shaded, wall, tree trunk), and then make a labelled drawing of one moss plant, with a title, magnification, and label lines for rhizoids, 'stem', 'leaf-like structures', stalk and capsule (if present).	groups to share textbooks. Ask 2 groups to report what they found most surprising.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy bryophyta moss liverwort  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy bryophyta moss liverwort	Groups display their labelled drawings and tables around the room for a gallery walk. Each learner visits at least three other groups' displays with sticky notes, leaving one 'Star' (something accurate or well labelled) and one 'Question' (something unclear). Back in their group, learners read the feedback and revise their drawings. Then, in Think-Pair-Share, they answer: 'Why does moss stay small and grow only in damp places? How does it reproduce without seeds or flowers?' Using their observations and the textbook, they write an explanation that lists the characteristics of Division Bryophyta: no vascular tissue (no xylem and phloem), rhizoids instead of true roots, thin leaf-like structures, spores produced in capsules, and need for water for fertilisation. They return to their predictions and mark them as supported, partly supported or changed.	Say: 'Leave your drawing on the desk and walk quietly. Read the labels and tables and give feedback using the Star and Question format.' Allow 8 minutes for the gallery walk. Then ask: 'Which features did every group find in moss?' WAIT TIME 10-15 seconds, then cold-call 4 learners from different groups. Record agreed characteristics on the board as 'Class evidence'. Pose: 'Maize carries water up through its stem. What does moss do without xylem and phloem? What does that tell us about its size and where it can live?' Give 2 minutes for Think-Pair-Share, then cold-call 3 pairs. Introduce key terms (Bryophyta, rhizoids, spores, capsule) only after learners describe the features in their own words. Address misconceptions: 'Rhizoids anchor the plant and do not transport water like true roots.' Link to Kenya: 'Where have you seen moss near home, on rocks near Kericho tea estates or damp walls in Nairobi? What is common about those places?'	Gallery walk with Think-Pair-Share: Peer feedback exposes differences in how groups interpret the same structures and pushes them to justify their labelling with evidence. Connecting structure (no vascular tissue, rhizoids) to function (small size, moist habitat) turns a list of facts into an explanation, making the classification meaningful instead of memorised.	Look for learners who: (1) explain moss's small size and damp habitat by referring to the lack of vascular tissue and the need for water; (2) correctly distinguish rhizoids from true roots; (3) revise their prediction with evidence. Use thumbs up/sideways/down on 'I can list three characteristics of bryophytes' and follow up with learners who show sideways or down. Collect the Star and Question sticky notes to identify common errors.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>	Learners return to the class Driving Question Board with the central question: 'How can we identify and classify living organisms accurately?' Each group writes on sticky notes (1) one thing they have learned about moss and bryophytes that helps classification and (2) one new	Say: 'Look at the Driving Question Board. Today we added evidence about one group of plants. What can we now say, and what do we still need to find out?' Give 2 minutes of silent writing, then WAIT TIME 10-15 seconds before inviting contributions. Cold-call 3 groups to read out and place one	Driving Question Board: Collecting learner-generated questions makes the unit's direction visible and shows that new evidence raises new questions. Sorting questions by type helps learners see how today's work on one plant group connects to the larger need for a standard classification	Check that sticky notes (1) state a learned idea supported by evidence from today (e.g. 'Moss has no true roots, only rhizoids'); (2) contain at least one researchable question; and (3) are linked to the driving question. Note learners who cannot yet generate questions, and pair them with

 Search ARES for videos Search terms: taxonomy bryophyta moss liverwort  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy bryophyta moss liverwort	question, for example 'Do ferns have roots like maize?', 'What makes bean plants different from moss?', 'How do plants with seeds reproduce?' They place notes on the board under headings such as 'What we learned', 'New questions' and 'Still not sure', and group similar questions with the teacher.	question each. Sort questions with the class: 'Which questions are about other plants? Which about animals? Which about viruses?' Highlight the question 'What makes other plants different from bryophytes?' and circle it as the focus for Lesson 3. Praise question quality: 'This question asks for evidence we can gather - that is a strong scientific question.' Ensure respectful tone: 'We value each person's ideas.'	system.	stronger peers next lesson.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: taxonomy bryophyta moss liverwort  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy bryophyta moss liverwort	Groups revisit their Lesson 1 classification model (chart of maize, bean, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat). They revise it by creating a 'Bryophyta' category for moss based on the shared observable features gathered today, and write the criteria beside it (e.g. 'small, no true roots, stem or leaves, rhizoids, no vascular tissue, spores in capsule, grows in damp places'). They mark where other organisms (maize, bean, grass, fern) still do not fit and need more evidence. Individually, learners complete an exit ticket summarising the lesson in the Observed / Learned / Explained table.	Say: 'Take out your Lesson 1 model. Using today's evidence, what must change? Add a group for moss and write the features that decide membership.' Circulate and ask: 'Which feature is the most useful for telling moss apart from maize? Why is colour not a good feature?' WAIT TIME 10-15 seconds. Cold-call 3 groups to share a revised part of their model, and ask others: 'Do you agree? What evidence supports that?' Point out that scientists in Kenya, such as botanists at the National Museums of Kenya herbarium, use shared characteristics like these to name and group plants. Close with: 'We now know what defines bryophytes. What do you think makes other plants, like ferns and maize, different? That is our question for the next lesson.' Distribute the exit ticket.	Model revision: Updating the classification model with evidence shows learners how scientific ideas change as data accumulate. Writing explicit criteria for a group moves learners from sorting by appearance or usefulness toward using shared, observable characteristics, which advances the unit's driving question.	Check that revised models (1) include a labelled Bryophyta group with at least three observable criteria; (2) avoid non-scientific criteria (size alone, colour, usefulness); and (3) show where unresolved organisms remain. Review exit tickets: successful responses cite at least two bryophyte characteristics and one evidence-based reason for moss's habitat. Use results to plan support for Lesson 3.

D. TEACHER REFLECTION

Which learners explained moss's small size and damp habitat using evidence (no vascular tissue, rhizoids) rather than just recalling facts? Did learners confuse rhizoids with true roots, or moss with fungi such as mushroom? Were specimens available, or did pictures limit observation of the capsule? How effective was the gallery walk in

improving drawing quality and the accuracy of labels? Did groups' revised models move from appearance or use toward shared structural characteristics? Which new questions on the DQB should drive the Lesson 3 evidence-gathering activity on ferns, conifers and flowering plants?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Describe what you observed in the moss with a hand lens: its size, the base (rhizoids), the stem-like and leaf-like parts, any spore capsules, where it was growing, and how it differed from maize or bean plants.
What did I learn?	State the characteristics that identify Division Bryophyta (e.g. small size, rhizoids instead of true roots, no vascular tissue, spores in capsules, damp habitats) and how they differ from other plants.
How does this explain the phenomenon?	Explain why moss stays small and grows in damp places, and how this evidence helps scientists classify plants more accurately than grouping by size, colour or use. What new question do you now have about how other plants differ from bryophytes?

LESSON 3 (80 minutes (2 x 40-minute periods)): Pteridophyta and Spermatophyta

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence from real plants (fern, maize, bean) to show that pteridophytes and spermatophytes have features that distinguish them from bryophytes and from each other, and use it to improve their class model of how organisms are grouped.
Knowledge	a) outline the general characteristics of organisms in Divisions Bryophyta, Pteridophyta and Spermatophyta. Pteridophytes (ferns) have true roots, stems and leaves (fronds) with vascular tissue and reproduce by spores in sori, with no seeds or flowers. Spermatophytes have true roots, stems and leaves with vascular tissue and reproduce by seeds (in cones or flowers). Bryophytes (moss) lack true roots and vascular tissue.
Skills	Observing and recording structures of roots, stems, leaves and reproductive parts; comparing and contrasting specimens; constructing a comparison table; (NGSS SEP) analysing and interpreting data, and developing and using models; Communication and Collaboration: speaking clearly to peers while discussing the distinguishing characteristics of various specimens.
Attitudes	e) appreciate the diversity of plants and animals in nature. Respect: considerate of others' opinions during group discussion. Responsibility: cares for the environment while collecting specimens. PCI Environmental Conservation: collects only small samples responsibly and leaves the habitat intact.
Key Inquiry Question	1. How are plants and animals classified?
Purpose in Storyline	Lessons 1 and 2 showed that learners' own groupings (size, colour, use) conflict, and that moss (Bryophyta) is grouped by structural features. In Lesson 3 learners see that features such as vascular tissue, spores versus seeds, and flowers separate more plant groups. This raises the next question: 'How can we classify plants systematically when we encounter an unknown plant?', which leads to the dichotomous key work in Lesson 4.
Safety Notes	Collect only small samples (one frond, one seedling) from the school compound, with teacher permission. Do not collect unknown plants, and do not taste any plant or mushroom. Wash hands after handling soil and plants. Avoid plants with thorns or irritant sap, such as euphorbia. Use hand lenses carefully and never look at the sun. Pull up bean or maize seedlings gently and return waste to the compost pit.

B. LESSON OVERVIEW

Back to our phenomenon: the maize plant, bean plant, grass, fern, moss, mushroom and other organisms from the school compound are all living, but our own groupings by size, colour or use disagree. Today learners focus on the plants. They predict how a fern differs from moss and from maize or beans, then examine real fern fronds (with sori on the underside), maize and bean seedlings or plants (or pictures where specimens are unavailable). Working in Jigsaw expert groups (roots, stems, leaves, reproductive structures), they gather evidence, return to home groups to build a comparison table of Bryophyta, Pteridophyta and Spermatophyta, then take a Gallery walk to compare tables and resolve differences. A short video (Video 2) or chart supports the observations but is not essential if there is no internet. Learners add questions to the Driving Question Board and revise their class model of plant grouping, ending with the new question: how do we classify an unknown plant systematically?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 05 Plant Diversity And Evolution, Seedless Vascular Plants (Ferns) <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy pteridophyta spermatophyta fern	Learners recall the compound organisms from the phenomenon (maize, bean, grass, fern, moss). On a card each writes a prediction: 'How will a fern differ from moss, and from a maize or bean plant? Think about roots, stems, leaves and how they make new plants.' They think alone for 2 minutes, then pair and share, drawing a quick sketch of what they expect to see on the underside of a fern frond. Kiswahili may be used for discussion; key terms are written in English.	Hold up a fern frond, a maize plant and a bean plant beside the moss from Lesson 2. Say: 'We said earlier that our own groups did not agree. Today we ask what features of these plants everyone could check.' Ask: 'Which of these has real roots? Which makes seeds? How do you know?' Give WAIT TIME of 10-15 seconds after each question before anyone speaks. Cold-call 4 learners to share predictions, and ask 2 of them: 'What makes you think so?' Record predictions on the board under 'Our Predictions' without correcting them. Remind learners: 'A prediction is our best idea, not a right or wrong answer.'	Think–Pair–Share (with the Predict stage of POE): individual thinking exposes prior ideas (for example 'all green plants make seeds', 'ferns are flowering plants'), and pairing lets learners test ideas aloud, so the investigation has a purpose.	Look for: (1) at least 3 of 5 learners' cards include a feature (roots, seeds, spores, flowers) rather than just size or colour; (2) learners give a reason with 'because'; (3) note common misconceptions on the board, such as 'ferns have flowers' or 'moss has roots'. Use a tally of misconceptions to revisit in the Explain phase.
Observe Phase  VIDEO: 05 Plant Diversity And Evolution, Seedless Vascular Plants (Ferns) <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES	Jigsaw, Part 1 (expert groups, 20 minutes): Class is split into 4 expert groups: Roots, Stems, Leaves, and Reproductive structures. Each group examines a fern, a maize plant and a bean plant (and the moss from Lesson 2 for comparison), using hand lenses and, where needed, drawings or pictures from the textbook or charts. Roots: fibrous (maize) vs tap root (bean) vs rhizome with fibrous roots (fern) vs rhizoids (moss). Stems: rhizome in fern, herbaceous stems with vascular bundles in maize and bean. Leaves: fronds with pinnae (fern); long narrow leaves with parallel veins (maize); broad leaves with net veins (bean). Reproductive: brown sori on the	Set up stations before class. Safety reminder: 'Handle plants gently, take only what I give you, and wash your hands afterwards.' Circulate and ask: 'What do you see on the underside of the fern frond? How is it different from the bean leaf?' and 'Where are the seeds in the bean? Where would you look for seeds in the fern?' Give WAIT TIME of 10-15 seconds after each question. Prompt without telling: 'Count the veins. Do they run parallel or form a net?' Cold-call 2 experts per group in Part 2 to explain their findings: 'Tell your group one thing you saw and one piece of evidence.' Make sure each group has one fully completed table. If Video 2 is used, play it once for 3-4 minutes,	Jigsaw cooperative learning: each learner becomes responsible for a piece of evidence and must explain it to others, so every learner contributes to the final table and builds understanding of the characteristics. Building a comparison table organises observations into patterns across groups.	Observable indicators: expert sketches show correct labels (for example sori, frond, tap root, fibrous roots); learners use words like 'spores' and 'seeds' in explanations; each home group table has all 4 features for all 3 divisions; learners can state at least one feature that separates ferns from moss (true roots, vascular tissue) and ferns from seed plants (spores, no seeds). Use a checklist during circulation.

for readings Search terms: taxonomy pteridophyta spermatophyta fern	underside of a fern frond (spores); maize cob and tassel (flowers, grains); bean flowers and pods with seeds. Learners record observations in labelled sketches and notes. Jigsaw, Part 2 (home groups, 15 minutes): Learners regroup so each home group has one expert from every topic; each expert teaches the others, and the group builds a comparison table: Feature Bryophyta (moss) Pteridophyta (fern) Spermatophyta (maize, bean). Where no fern is available, use charts or Video 2, and pause the video at key points for learners to record notes.	say: 'Watch for spores and seeds. Note down two things you did not see in the specimens,' then pause to share in pairs.		
Explain Phase  VIDEO: 05 Plant Diversity And Evolution, Seedless Vascular Plants (Ferns) <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy pteridophyta spermatophyta fern	Gallery walk (12 minutes): Each home group posts its comparison table on the wall. Groups rotate every 2 minutes, using sticky notes to add a star (a strong piece of evidence), a question mark (unclear), or a suggestion. Back at their own tables, groups revise them. Then in a class discussion learners revisit predictions and answer: 'What features separate Pteridophyta from Bryophyta? What separates Spermatophyta from Pteridophyta?' They explain using evidence: ferns have vascular tissue, true roots and fronds but reproduce by spores in sori; spermatophytes have vascular tissue and reproduce by seeds (in flowers or cones). They notice maize and bean differ too (parallel vs net veins, fibrous vs tap root), hinting at monocots and dicots for later. Learners write 2-3 sentences in their notebooks: 'The feature that best separates a fern from a bean plant is ____ because ____.'	Say: 'Walk like scientists. Look for evidence, not just opinions. Where do tables agree? Where do they disagree?' Ask during the walk: 'Which feature appears in every table? Why would scientists use this feature rather than colour or size?' Give WAIT TIME of 10-15 seconds. Cold-call 5 learners after the walk: 'What did you change in your table and why?' Probe the misconceptions from the Predict phase: 'Earlier some of us said ferns have flowers. What does the evidence say?' Introduce terms only after learners use their own words: spores, sori, vascular tissue, fronds, seeds. Link back to the phenomenon: 'Remember our groups by size or by use as food? Would a farmer in Kericho and a student in Kisumu both be able to check spores and seeds?' Mention Kenyan examples: bracken fern in Aberdare forests, maize and beans (githeri) as spermatophytes, and Kenyan scientist Prof. Wangari Maathai's tree planting for conservation (trees are	Gallery walk with Claim–Evidence–Reasoning prompts: comparing the work of other groups exposes differences, makes learners justify claims with observed evidence and builds consensus on shared, observable characteristics, which is the basis of classification.	Check notebook sentences: look for (1) a correct feature, (2) evidence from specimens, (3) a reason linking to classification. Success criteria: at least 80% of learners name spores vs seeds correctly. Look for sticky notes that give constructive feedback. Identify learners who still confuse ferns with flowering plants for follow-up with a diagram.

		spermatophytes).		
Driving Question Board (DQB) Creation  VIDEO: 05 Plant Diversity And Evolution, Seedless Vascular Plants (Ferns) <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy pteridophyta spermatophyta fern	Learners return to the Driving Question Board with the question 'How can we identify and classify living organisms accurately?' They write new questions on sticky notes based on today's work, for example: 'Why do ferns need damp places?', 'How do we tell a maize plant from grass?', 'What about plants with cones, like cypress?', 'How do we classify a plant when we do not know it?' They place the notes under categories (Plants, Animals, Method, Viruses) and mark one that they think can be answered next lesson. They also move or tick questions answered today, such as 'How is moss different from a fern?'	Say: 'Look at the board. Which of our questions did today's evidence help to answer? What new questions came up?' Give WAIT TIME of 10-15 seconds, then ask each pair to write at least one question. Cold-call 3 pairs to read out and place questions, asking: 'Why does that question belong there?' Highlight the question that links to the next lesson: 'How can we classify plants systematically when we encounter an unknown plant?' Praise questions that need evidence and not just yes/no answers. Accept Kiswahili questions and translate key terms into English with the class.	Driving Question Board: it makes learners' thinking visible, connects each lesson to the anchoring phenomenon, and shows progress and gaps, driving the need for the next investigation.	Look for: each pair contributes at least one question; at least half the questions refer to features or methods (not just facts); some questions show links between divisions or ask about unknown plants. Record question types (what, why, how) to plan the next lesson, and tick off answered questions.
Model Building Phase  VIDEO: 05 Plant Diversity And Evolution, Seedless Vascular Plants (Ferns) <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading</i>	Learners revise the class model (a flow chart or concept map) that shows how organisms from the school environment can be grouped. They add Pteridophyta and Spermatophyta next to Bryophyta, and draw the features used at each branch, for example: Vascular tissue? No: Bryophyta (moss). Yes: Seeds? No, spores: Pteridophyta (fern). Yes, seeds: Spermatophyta (maize, bean). In groups, they add the maize, bean, grass, fern and moss from the phenomenon into the model and note which ones are still unclear (for example mushroom, which is not a plant). Each learner	Say: 'Our first groups did not agree. Now, using features anyone can observe, where do our plants go? Show me on the model.' Draw the model skeleton and let groups add labels. Ask: 'What evidence supports this branch? What is missing?' Give WAIT TIME of 10-15 seconds. Cold-call 4 learners to place one organism and justify it. Ask: 'Where would the mushroom go? Why not here?' (leave unresolved; it is revisited in Lesson 8). Close with: 'If a farmer in Rift Valley brings you an unknown plant, what would you check first? How could we make this a step-by-step tool?' and introduce Lesson 4:	Model revision: updating the class model with new evidence helps learners organise features into a hierarchy and see why a standard, shared system works better than personal groupings, and it exposes the limits that lead to dichotomous keys.	Observable indicators: the model correctly places moss, fern, and maize/bean using spores vs seeds and vascular tissue; learners justify at least one placement with evidence; the exit ticket names one limitation of the model. Use a 3-level rubric (placement correct, evidence stated, limitation identified) to decide who needs extra support with diagrams or reteaching before Lesson 4.

<i>in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy pteridophyta spermatophyta fern	completes a summary table in their notebook (What we observed / What we learned / How it helps explain the phenomenon) and notes one limit of the model: 'It does not yet tell us how to identify an unknown plant step by step.'	dichotomous keys. Collect model sketches as an exit ticket.		
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D. TEACHER REFLECTION

Did every learner handle real specimens or clear pictures and contribute to the Jigsaw? Which misconceptions (ferns have flowers, moss has roots, all plants have seeds) remained after the Gallery walk? Was WAIT TIME of 10-15 seconds used consistently, and did quieter learners speak when cold-called? Were specimens (ferns) available locally, or did I rely on pictures, and how did this affect learners' evidence? Did learners connect their table back to the phenomenon and our conflicting groupings? What should I adjust for Lesson 4 (dichotomous key challenge) based on the DQB questions?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that ferns have true roots, a rhizome and fronds with brown sori on the underside, while maize and bean plants have roots, stems, leaves and flowers that produce seeds (grains and pods). Moss has no true roots and no vascular tissue.
What did I learn?	Pteridophytes reproduce by spores and have vascular tissue but no seeds or flowers; spermatophytes reproduce by seeds and have vascular tissue; bryophytes lack vascular tissue and true roots. These are shared, observable features anyone can check.
How does this explain the phenomenon?	These features help explain our phenomenon because they give everyone the same rules for grouping plants, instead of personal groupings by size, colour or use as food. We still need a step-by-step way to identify an unknown plant.

LESSON 4 (80 minutes (2 x 40-minute periods)): Using Dichotomous Keys

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners figure out that a dichotomous key uses pairs of contrasting, observable characteristics to identify organisms. They use this idea to improve the school-organism classification models they started in earlier lessons, so that anyone can identify an organism and get the same answer.
Knowledge	By the end of the lesson the learner should be able to: (1) construct and use dichotomous keys to identify various plants (KICD: construct and use dichotomous keys to identify various plants and animals); (2) state the principles of constructing a dichotomous key: paired contrasting statements, observable features, one clear choice at each step, numbered couplets; (3) relate the key to the distinguishing features of Bryophyta, Pteridophyta and Spermatophyta, and of monocots and dicots.
Skills	NGSS SEP: Developing and Using Models (revising the group classification model into a key); Analyzing and Interpreting Data (comparing the results of different users of the same key); Constructing Explanations (explaining why a key gives the same answer for every user). Core competencies: Communication and Collaboration (speaking clearly to peers while discussing distinguishing characteristics); Creativity and Imagination (trying varied ideas while designing the key).
Attitudes	Values: Respect (being considerate of others' opinions when groups disagree on a feature) and Responsibility (caring for the environment while collecting plants). PCI: Environmental Conservation (collecting only small samples responsibly and leaving the habitat undisturbed). Appreciating the diversity of plants in the local environment.
Key Inquiry Question	KICD Key Inquiry Question 1: How are plants and animals classified? Driving question: How can we identify and classify living organisms accurately?
Purpose in Storyline	In Lessons 1-3 learners saw that their own groupings (size, colour, habitat, usefulness as food) differ between groups, and they met the shared characteristics of plant divisions. Lesson 4 gives them a tool that turns those observable characteristics into a step-by-step identification method that any person gets the same answer from. It ends with the question: Can the same method be used to identify animals? This leads to Lesson 5.
Safety Notes	Collect only plants that are common and known (maize, bean, grass, fern, moss, sukuma wiki, Mexican marigold). Avoid unknown plants, thorny or irritating plants (e.g. nettle), and wild mushrooms; mushrooms are used only as pictures. Wash hands after handling soil and plants. Take only one leaf or small sample per plant; do not uproot plants. Move in groups, stay within the school compound. Use rulers carefully.

B. LESSON OVERVIEW

Learners return to the school-organism phenomenon: 'If all these organisms are living things, how can scientists organise them so that any person anywhere can identify them?' Groups first predict whether two different people could use the same set of written steps to name the same plant. They then study a sample key, use it on local plants (maize, bean, grass, fern, moss, sukuma wiki leaf) and see that different groups reach the same answer, which is different from their earlier personal groupings. In the Explain phase they use Think-Pair-Share to figure out the rules of a good key and complete the Dichotomous-key Challenge of building a key for their own collected plant specimens. They add the new questions to the Driving Question Board and revise their classification model from earlier lessons. Activities need no internet. Prepared keys, specimens and worksheets are used; photographs or drawings replace any missing specimen. Discussion may begin in Kiswahili or mother tongue, while scientific terms are recorded in English.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key contrasting	(10 min) Learners sit in their groups of 4-5 with the group classification charts they made in earlier lessons (grouped by size, colour, habitat, food use). The teacher holds up a maize plant and a fern and asks: 'Imagine a farmer in Kericho wants to name this plant, but you are not there. Could you write instructions so that she gets the same name you do?' Individually, learners write a prediction in their notebooks: 'I think instructions for identifying plants should look like...' and sketch or list the first two steps they would write. They then share in pairs (Think-Pair-Share), using Kiswahili if helpful, and record a short agreed prediction on the group chart: 'Our key will work if...'	Display the maize plant, fern, bean plant and moss on the front table. Say: 'In Lesson 1 your groups sorted these organisms in different ways, and the groups did not agree. Today we ask: how can any person anywhere get the same name for the same organism?' Give 1 minute silent think time. WAIT TIME: 10-15 seconds after asking 'What would make the instructions work for everyone?' before taking any answer. Cold-call 3 pairs (not volunteers) to read their predictions aloud. Say: 'I am not looking for the right answer yet; I want your best prediction.' Write 3 predictions on the board, circle common words (e.g. 'leaf', 'flower', 'yes/no'). Do not reveal the term dichotomous key yet.	Predict-Observe-Explain (the Predict step) combined with Think-Pair-Share. Predicting before instruction exposes learners' preconceptions (e.g. that grouping by colour or size is enough) and gives a baseline that learners will compare with evidence later. Think-Pair-Share gives every learner thinking time and a chance to rehearse ideas before whole-class talk.	Look for: (1) predictions that mention observable features (leaf shape, presence of flowers, seeds, roots) rather than opinions (useful, pretty); (2) at least 3 of the cold-called pairs mention that steps must be clear or have only two choices; (3) number of learners who still suggest size or colour as the only basis. Tally on a class list: 'observable feature / opinion / no response'. Note groups who need support at the Observe stage.
Observe Phase  VIDEO: 25 Dichotomous Key I Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key contrasting	(25 min) Part A (8 min): Each group receives a prepared sample key for six local plants, worksheets and rulers. They read the first couplet together (e.g. 1a: Plant has no true roots, stems or leaves; 1b: Plant has true roots, stems and leaves), and the teacher models it with moss. Part B (17 min): Groups go outside or to a specimen table (plant specimens, photographs or drawings where plants are missing) with maize, bean plant, grass, fern, moss and sukuma wiki. Using the key, a ruler and observation, they identify each unknown specimen labelled A-F, recording the path taken through the couplets (e.g. 1b to 2a to 3b) and the final name. They measure features that need numbers (e.g. leaf length, number of leaf veins or	Model the first couplet aloud with moss: 'Does it have true roots, stems and leaves? Look carefully, then choose one statement.' Circulate and ask each group: 'Which two statements are you choosing between? What do you see that tells you?' WAIT TIME: 10-15 seconds after each question before prompting. Avoid telling them the plant name; instead say 'Show me your path through the key.' Cold-call 2 learners per group to explain one choice. Remind learners: 'Collect only a small piece; do not pull up the whole plant', and 'Wash your hands afterwards.' Ask groups that finish early: 'Where did your partner choose differently from you? Why?' Use photographs for any plants unavailable, and tell	Predict-Observe-Explain (the Observe step) with the Dichotomous-key Challenge. Using a ready-made key on real local specimens lets learners experience that the same steps give the same answer, which directly tests their earlier prediction and shows the difference from their own unstructured groupings. Paired verification builds evidence from repeated trials.	Observable indicators: (1) group records a complete path through the key for at least 5 of 6 specimens; (2) learners point to the actual feature on the specimen when justifying a choice; (3) rulers are used correctly (zero-end alignment, units recorded); (4) the second user of the key reaches the same name for at least 4 of 6 specimens. Use a checklist while circulating. Record which couplets caused disagreements for use in the Explain phase.

	petals) with rulers. Groups swap and a second member repeats the key on one specimen to see whether the answer matches. They note disagreements and which statement caused them.	learners that mushrooms are observed from pictures only.		
Explain Phase  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key contrasting	(25 min) Step 1 (8 min) Think-Pair-Share: Learners compare results with their prediction and answer: 'Why did the key give the same answer to different people when our earlier groupings did not?' Step 2 (7 min): Pairs study the sample key and list what makes each couplet good (two contrasting statements, observable features, one choice only, no opinions such as 'useful' or 'beautiful', numbered). Whole class builds a shared list of principles of construction. Step 3 (10 min) Dichotomous-key Challenge: Each group takes 4 plants they collected or were given (e.g. maize, bean, fern, moss, grass or sukuma wiki) and writes a simple key with at least 3 couplets using visible features (e.g. parallel veins versus net veins, fronds versus broad leaves, presence of seeds). They swap keys with another group, which tries to identify the specimens with it and gives feedback: 'Which couplet was unclear?' Groups then revise their key. Learners link their features to the divisions: moss (Bryophyta), fern (Pteridophyta), maize and grass (monocots), bean and sukuma wiki (dicots) in Spermatophyta.	Ask: 'Why did the key give the same answer to different people, when your earlier groupings did not?' WAIT TIME: 10-15 seconds, then cold-call 4 pairs. Press for reasoning: 'Which feature was a better choice than colour? Why?' Challenge weak couplets: 'Is that statement something we can see and measure, or is it an opinion?' Write learners' principles on the board and add missing ones with the KICD wording (principles of construction, identification of organisms). During the challenge, circulate with the prompt: 'Could someone from Mombasa who has never seen this plant use your key?' Build in respect: remind groups to give feedback kindly and to listen to others' ideas ('First say one thing that works well, then one suggestion'). Connect back to the phenomenon: 'Our school organisms are all living things, yet this method lets anyone name them.'	Think-Pair-Share, then a Dichotomous-key Challenge with peer testing. Explaining why the key works converts observations into an understanding of the concept (contrasting pairs, observable features). Constructing a key for their own specimens and having another group test it applies the concept and uses peer feedback to build accuracy, in line with the NGSS practice of constructing explanations and revising models.	Indicators: (1) learners state that a good couplet contains two contrasting, observable statements; (2) keys have at least 3 numbered couplets that lead to a single name for every specimen; (3) the swapping group correctly identifies at least 3 of 4 specimens; (4) learners correctly link at least two features to the division or class (e.g. parallel veins = monocot; spores/fronds = fern). Use a quick rubric (3 = key works without help, 2 = one unclear couplet, 1 = key needs reteaching). Misconceptions to note: using size/colour as the only basis, or writing statements with more than two choices.
Driving Question Board (DQB) Creation  VIDEO:	(10 min) Learners return to the class Driving Question Board headed 'How can we identify and classify living organisms accurately?'. On sticky notes, each	Say: 'Choose one question that you think our next lesson should investigate.' WAIT TIME: 10-15 seconds of silent writing before notes go up. Cold-call 3 groups to	Driving Question Board. The DQB makes learning visible over the unit, so learners see how today's evidence answers earlier questions and generates new	Indicators: (1) each group posts at least one 'what we learned' and one new question; (2) at least half of the questions refer to animals or exceptions to a key; (3) learners

25 Dichotomous Key I Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key contrasting	group writes (a) one thing they now understand about identifying plants (e.g. 'A key uses pairs of opposite features'), and (b) one new question (e.g. 'Can the same method be used to identify animals?', 'What if an organism fits neither choice?', 'How do we key organisms such as a mosquito or an earthworm?'). Groups place notes under the headings 'Answered', 'Partly answered' and 'New questions' and read other groups' notes during a short walk around the board.	read one note and explain where it belongs. Move answered notes only after asking the class: 'What evidence from today shows this question is answered?' Highlight the question 'Can the same method be used to identify animals?' as the bridge to Lesson 5, and link it to the phenomenon: 'We have a method for plants in our school, but what about the mosquito, butterfly, earthworm, snail, chicken, lizard and goat?' Acknowledge and value all questions respectfully.	ones. This keeps the unit phenomenon-driven by letting students' questions guide the next investigation.	can justify the position of a note using evidence from the key activity. Record questions that reveal confusion (e.g. 'Is a key the same as a table of groups?') for follow-up.
Model Building Phase  VIDEO: 26 Dichotomous Key II Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key contrasting	(10 min) Groups take out their classification model from Lessons 1-3 (chart or diagram of the school organisms, including their original grouping and any added division/class features). They revise it by (1) adding the dichotomous key they built as a 'tool for identification' alongside their groups; (2) crossing out or relabelling groupings based on opinion or unreliable features (e.g. 'good to eat'); (3) adding arrows linking observable features to the groups (Bryophyta, Pteridophyta, monocots, dicots); and (4) writing a short caption: 'Our model now explains that organisms can be identified accurately because...'. They mark with a different colour what changed and why, and leave a box labelled 'Animals: still to be tested'.	Say: 'Look at your model from last lesson. What can it now explain that it could not before? What still does not work?' WAIT TIME: 10-15 seconds before groups begin. Circulate and ask each group: 'Which piece of evidence from today made you change this part of the model?' Cold-call 2 groups to share one revision. Prompt: 'Does your model show why two people using the same key get the same name?' Collect models for display and use in the Lesson 8 final model. Link ahead: 'Next lesson we ask whether the same method works for the animals on our school list.'	Model revision. Updating the model with new evidence helps learners see classification as a developing explanation, not a list of facts. Colour-coding revisions makes their thinking visible and gives evidence of conceptual change.	Indicators: (1) the model shows the key as a step-by-step tool and not just groups; (2) at least two revisions are tied to evidence (a feature or result from the activity); (3) captions use terms such as contrasting, observable, couplet, monocot, dicot; (4) a 'still to be tested' box for animals is present. Use a 3-point rubric: complete revision with evidence, partial revision, no revision.

D. TEACHER REFLECTION

After the lesson ask yourself: Did learners use observable features (leaf veins, fronds, seeds) rather than opinions when making choices? Which couplets caused disagreements, and why? Did every learner (including those who used Kiswahili first) contribute to building the key? Were specimens, photographs or drawings enough for all groups? Did collection of plants stay responsible and minimal? Which learners still rely on size or colour to classify, and how will you support them in Lesson 5

when keys are applied to animals? Record the questions on the DQB that will guide the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We used a sample dichotomous key and a ruler to identify local plants (maize, bean, grass, fern, moss, sukuma wiki) and found that different group members reached the same name when they followed the same couplets.
What did I learn?	A dichotomous key uses pairs of contrasting, observable characteristics, in numbered couplets, so that each step has only two choices and leads to one identification. Features such as true roots, stems and leaves, fronds, seeds, and parallel versus net veins help place plants into Bryophyta, Pteridophyta and Spermatophyta (monocots and dicots).
How does this explain the phenomenon?	A key gives the same answer to everyone because it uses clear, observable features rather than personal opinions such as size, colour or usefulness. This is how scientists can organise the organisms around our school so that anyone can identify them. New question: can the same method be used to identify animals?

LESSON 5 (80 minutes (2 x 40-minute periods)): Kingdom Animalia — major phyla

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use observable body features of animals from the school environment to group them into major phyla and justify their placement, advancing the unit question: How can we identify and classify living organisms accurately?
Knowledge	Specific learning outcome: By the end of the sub strand, the learner should be able to: c) classify organisms into phyla and classes of Kingdom Animalia. Learners recognise that animals are grouped into phyla using body features such as symmetry, body segmentation, presence of a backbone, type of body covering, and jointed limbs. They link Annelida (earthworm), Mollusca (snail), Arthropoda (mosquito, butterfly) and Chordata (chicken, lizard, goat) to the organisms in the phenomenon.
Skills	Science and Engineering Practices: Analysing and interpreting data (recording body features in a table); Constructing explanations using Claim-Evidence-Reasoning; Engaging in argument from evidence (justifying phylum placement); Developing and using models (revising the class classification model). Core competencies: Communication and Collaboration - the learner speaks clearly to peers while discussing distinguishing characteristics of various specimens. Creativity and Imagination.
Attitudes	Values: Respect - the learner is considerate of others' opinions while discussing groupings. Responsibility - the learner cares for the environment while collecting specimens. PCI - Animal Welfare: the learner handles animals with care while observing and grouping them using distinguishing characteristics. PCI - Environmental Conservation. Appreciate the diversity of plants and animals in nature.
Key Inquiry Question	KICD key inquiry question: How are plants and animals classified? Lesson question: What body features allow us to place the animals we found around our school into major phyla, and how do we defend our choices with evidence?
Purpose in Storyline	Lessons 1-4 showed that groupings by size, colour or usefulness differ between groups and that scientists use shared, observable characteristics, including Kingdom Plantae divisions. Lesson 5 turns to the animals in the phenomenon (mosquito, butterfly, earthworm, snail, chicken, lizard, goat). It gives learners the evidence that animals can be sorted by body plan and leads to the next question: How are animals within the same phylum further classified?
Safety Notes	Handle live animals gently and return them to where they were found (Animal Welfare). Do not collect or touch unknown, venomous or dangerous organisms such as snakes, scorpions, bees or wasps, or stinging caterpillars. Use preserved specimens, dead insects or pictures where live ones are not available. Wash hands with soap after handling soil, earthworms, snails or specimens. Wear gloves for snails and earthworms if available. Do not taste anything. Keep mosquitoes as pictures or dead specimens only. Return all organisms to the school garden after observation.

B. LESSON OVERVIEW

Learners return to the anchoring phenomenon: the maize, beans, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat around the school. After Lessons 1-4 they know groupings by colour or size disagree and scientists use shared observable features. In this lesson they focus on the animals. They first predict how they would group the eight animals, then observe specimens or pictures (earthworm, snail, grasshopper or butterfly, mosquito, tilapia or omena from Lake Victoria, lizard, chicken, goat) in expert Jigsaw groups looking at body symmetry, segmentation, shell, exoskeleton, jointed legs, backbone and body covering. They then return to home groups, teach each other, and complete a Gallery Walk of grouped specimens. They write Claim-Evidence-Reasoning statements to justify phylum placement, add new questions to the Driving Question Board, and revise their class classification model. The lesson ends with the question: if the goat and the chicken are both Chordata, how do we tell them apart? This leads to Lesson 6. Learners may discuss in Kiswahili or their mother tongue, while scientific terms are recorded in English. The lesson works without internet; the video is optional.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy kingdom animalia phylum  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: taxonomy kingdom animalia phylum	Learners recall the school-yard organisms from the phenomenon. In groups of 4 they receive picture cards or real safe specimens of the animals only: earthworm, snail, mosquito, butterfly, chicken, lizard, goat, plus tilapia or omena. Individually they write a prediction: 'I think the animals can be grouped into ___ groups because ____.' Then they Think-Pair-Share with a partner and sort the cards into their own groups on a sheet of manila paper, labelling the rule they used. Sorting rules are recorded on the board. Learners may use Kiswahili to discuss, e.g. 'wanyama wenye mgongo' and 'wasio na mgongo'.	Say: 'Remember our puzzle: if all these are living things, how can anyone anywhere organise them so that they can be identified? Today we look only at the animals.' Display the picture cards. Ask: 'Before we learn any new names, how would YOU group these eight animals? Write your prediction silently.' WAIT TIME 10-15 seconds before pairs talk. Circulate and listen for rules such as size, habitat, 'those we eat', 'those that fly'. Cold-call 4 groups to share their rules and write them on the board without judging. Ask: 'Did every group choose the same rule? What problem does that create if a scientist in Nairobi and one in Kisumu need to agree?' Put the sorted sheets aside for the Gallery Walk later.	Predict-Observe-Explain (POE) and Think-Pair-Share: writing a prediction before seeing evidence surfaces learners' existing ideas (size, habitat, food use) so they can later compare these with the scientific features. Think-Pair-Share gives every learner a voice, including those who speak in Kiswahili first.	Observable indicators: (1) Each learner has a written individual prediction with a reason. (2) Groups can state their sorting rule aloud. (3) At least 3 different rules are recorded on the board, showing the problem of non-standard grouping. (4) Learners use words like 'has a backbone' or 'has a shell' unprompted: note who does so. Tally learners who group only by habitat or food use to target in the next phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: taxonomy kingdom animalia phylum  READING: No closely matching reading in the ARES library for this activity. Search ARES	Jigsaw. Learners form home groups of 5 and each member joins an expert group focusing on one set of specimens or pictures and charts: Expert A: Annelida and Mollusca (earthworm, snail); Expert B: Arthropoda - insects (butterfly, mosquito, grasshopper); Expert C: Arthropoda - others (spider, millipede, crab or prawn picture from Mombasa); Expert D: Chordata - fish and reptiles (tilapia or omena, lizard); Expert E: Chordata - birds and mammals (chicken, goat). Each expert group observes the specimens/pictures, uses textbooks and charts, and fills a table: body symmetry, body segmentation, body covering (skin,	Set up stations before the lesson with specimens or labelled pictures. Remind about safety: 'Touch gently, wash hands, and return live animals to the garden.' Say: 'You are the expert. Your job is to find features that anyone, anywhere, could observe and check, not opinions like beautiful or useful.' Give 15 minutes for the expert groups. Circulate with questions: 'What do you see on the outside of its body? Does it have a backbone you can feel or see on the chart? How many legs, and are they jointed?' Prompt: 'Which feature would help us tell the snail from the earthworm?' WAIT TIME 10-15 seconds after	Jigsaw cooperative learning: each learner becomes responsible for one piece of the evidence, so all must communicate clearly and listen to peers (Communication and Collaboration). Using the same set of observable features for every animal supports the idea of a standard, shared system for classification.	Observable indicators: (1) Each expert table has at least 4 completed feature columns for each specimen. (2) Labelled sketches show key features (e.g. segments of earthworm, shell of snail, 3 body parts and 6 jointed legs of insect, backbone of fish/lizard/goat). (3) In home groups, each learner teaches without reading word for word. (4) The master table is completed by all members. Use a checklist to record who can correctly say whether each animal has a backbone and an exoskeleton. Note misconceptions such as 'a worm and a snake are the same' or 'a butterfly is not an animal'.

for readings Search terms: taxonomy kingdom animalia phylum	shell, exoskeleton, scales, feathers, hair), number and type of limbs, presence of a backbone, movement. They draw one labelled sketch. If available, a short video clip of animal phyla is used after observation, and groups can use textbook pages instead if there is no internet. Experts then return to home groups and each teaches their organisms' features using their table and drawing, while others add the information to a master table.	each question. If a group is stuck, point to the textbook page on phyla features. Cold-call 2 experts per station to name one feature they will teach. In the home groups, give 4 minutes per expert and then announce 'Switch!'. Monitor that every learner speaks. Show the video (5 minutes) only if available; pause twice to ask 'Which feature just mentioned matches our specimens?'		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy kingdom animalia phylum  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: taxonomy kingdom animalia phylum	Home groups use the master table to place each animal in a phylum: Annelida (earthworm), Mollusca (snail), Arthropoda (mosquito, butterfly), Chordata (chicken, lizard, goat, tilapia). They write a Claim-Evidence-Reasoning (CER) statement for two animals of their choice, e.g. Claim: 'The earthworm belongs to Phylum Annelida.' Evidence: 'Its body is soft, long and divided into many similar rings (segments) and it has no legs or backbone.' Reasoning: 'Annelids are classified by having a segmented body, so the earthworm fits this phylum.' They also explain why the mosquito and the goat are in different phyla even though both can bite or feed on blood or grass. A Gallery Walk follows: groups display their re-sorted specimen cards with phylum labels and CER statements, and visitors leave a sticky note 'I agree because...' or 'I disagree because...' with a question. Groups compare their new scientific grouping with their Predict Phase groupings and discuss what changed.	Say: 'You now have evidence. Scientists do not just say where an animal belongs, they defend it. Use the sentence frames: My claim is..., My evidence is..., This shows... because...' Model one CER on the board using the snail. Give 8 minutes for writing. Circulate and press: 'Is that feature observable? Could a learner in Mombasa check it?' WAIT TIME 10-15 seconds for thinking. Start the Gallery Walk for 10 minutes with a rotation signal every 2 minutes. Ask visitors to give feedback respectfully (Respect). After the walk cold-call 5 learners: 'Which group's reasoning convinced you, and what evidence made you change your mind?' Then ask: 'Compare today's groups with your first groups. Which rules were not useful for everyone?' Highlight that body structure features give the same result no matter who uses them. Introduce terms phylum, Annelida, Mollusca, Arthropoda, Chordata after learners have used the features, not before.	Claim-Evidence-Reasoning (CER) with Gallery Walk: CER links each claim to observed features and to the classification principle, building scientific argumentation. The Gallery Walk exposes learners to other groups' reasoning and encourages respectful peer critique, which resolves the disagreement seen in the Predict Phase.	Observable indicators: (1) CER statements name a phylum, cite at least two observed features and link them to the group characteristic. (2) Learners distinguish observable evidence from opinion. (3) Sticky-note feedback is specific and respectful. (4) Learners correctly explain why the earthworm (Annelida) differs from the snail (Mollusca) and why the mosquito and butterfly are both Arthropoda. Collect two CER statements per group for review; common errors (e.g. grouping by habitat) are revisited at the start of Lesson 6. Use a 3-question exit check: Which phylum has a backbone? Which has jointed legs and an exoskeleton? Which has segmented rings and no legs?
Driving	Learners revisit the Driving	Say: 'Look back at our board.	Driving Question Board: it makes	Observable indicators: (1) At least

Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy kingdom animalia phylum  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy kingdom animalia phylum	Question Board: 'How can we identify and classify living organisms accurately?' In groups they write new questions or answers on sticky notes, e.g. 'If the goat and chicken are both Chordata, how are they different?', 'Why is a bat a mammal and not a bird?', 'How are insects different from spiders?', 'Are all Arthropoda harmful, like mosquitoes?'. They place notes under categories: Answered today, Partly answered, New questions. Learners move one earlier question to 'Answered' with evidence from today's table.	Which of our questions can we answer with today's evidence? What new questions do the animals raise?' Give 3 minutes for silent writing, WAIT TIME 10-15 seconds before sharing. Cold-call 4 learners to read a question and place it on the board. Ask: 'Which question shows we need to look more closely within one phylum?' Highlight the question about how animals within the same phylum are classified. Link to the next lesson: 'All the Chordata we found share a backbone, yet a chicken, a lizard and a goat are very different. How do we tell them apart?' Praise questions that refer to the phenomenon organisms.	learners' growing understanding visible, connects each lesson to the anchoring phenomenon, and shows how evidence generates new questions that drive the next lesson.	one new question per group that refers to a specific organism. (2) Learners sort questions as answered or unanswered with a reason. (3) Some questions move beyond phylum level to classes within a phylum, showing readiness for Lesson 6. Note learners who still only ask naming questions (What is this called?) and encourage how/why questions.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy kingdom animalia phylum  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: taxonomy	Groups revise the class classification model started in earlier lessons. On chart paper they build or update a concept map/tree diagram with Kingdom Animalia at the top branching into the phyla Annelida, Mollusca, Arthropoda and Chordata, with the school organisms (earthworm, snail, mosquito, butterfly, chicken, lizard, goat) placed at the branches and the distinguishing feature written on each branch (e.g. 'segmented body, no legs', 'soft body with shell', 'jointed legs and exoskeleton', 'backbone'). They add Kingdom Plantae divisions from earlier lessons beside it and a note: 'Sorting rule must use observable features that everyone can check.' In a Think-Pair-Share, each learner finishes the sentences in the Summary	Say: 'Return to our first model. What would you add, change or remove after today's evidence?' Provide chart paper and markers; allow 12 minutes. Circulate and ask: 'Which feature on this branch separates these animals from the others? Could a visitor from Nairobi use your model to identify the chicken?' WAIT TIME 10-15 seconds. Cold-call 3 groups to explain one revision and the evidence that caused it. Remind learners to handle returned organisms carefully and release them in the garden (Animal Welfare). Close with: 'We can now say that animals are placed in phyla using body features. Next time we ask how animals within the same phylum are further classified.' Display models for the next lesson.	Model revision: updating a visible model with colour-coded changes helps learners see how evidence changes their thinking, linking the phenomenon of different groupings to a standard classification system based on shared features.	Observable indicators: (1) The model shows Kingdom Animalia branching to at least four phyla with distinguishing features. (2) All seven phenomenon animals are correctly placed. (3) Changes from the first model are marked and explained. (4) Learners can explain orally why a feature-based system works for everyone. Use the model and summary table as an exit ticket; learners who place organisms by habitat or use are given an extra pair of specimens to re-sort at the start of Lesson 6.

kingdom animalia phylum	Table prompt. Groups use a different colour to mark what changed from their first model.			
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D. TEACHER REFLECTION

Which learners still used size, habitat or food use rather than body features, and what evidence showed the change in their thinking? Did every Jigsaw expert have enough specimens or pictures and textbook access, and what substitutes worked where live animals were missing? Which CER statements lacked observable evidence, and how will I address that early in Lesson 6? Did learners use Kiswahili or mother tongue productively before recording English scientific terms? Were safety and animal welfare rules followed, and were all organisms returned to the garden? Which new DQB questions show readiness to explore classes within Chordata and Arthropoda?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the earthworm has a soft segmented body with no legs, the snail has a soft body with a shell, the mosquito and butterfly have jointed legs and a hard outer covering (exoskeleton), and the chicken, lizard and goat all have a backbone but different body coverings (feathers, scales, hair).
What did I learn?	We learned that animals are classified into major phyla (Annelida, Mollusca, Arthropoda, Chordata) using observable body features such as segmentation, shell, jointed legs with an exoskeleton, and a backbone, not by size, colour, habitat or usefulness.
How does this explain the phenomenon?	We can explain that the school animals can be organised so anyone can identify them because phylum placement is based on shared, checkable body features. For example, the goat and the chicken are both Chordata because they both have a backbone, while the mosquito is Arthropoda because it has jointed legs and an exoskeleton. We still wonder how animals within the same phylum, such as the chicken, lizard and goat, are further classified.

LESSON 6 (80 minutes (double lesson)): Classes of Kingdom Animalia

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use observable features (body covering, limbs, respiration and reproduction) to divide animal phyla into classes, then defend their classifications with evidence. This advances the driving question 'How can we identify and classify living organisms accurately?' by showing that accurate classification needs finer, shared characteristics than size, colour or habitat.
Knowledge	b) classify organisms into phyla and classes of Kingdom Animalia. Phylum Chordata is divided into Pisces, Amphibia, Reptilia, Aves and Mammalia. Phylum Arthropoda is divided into Insecta, Arachnida, Crustacea and Myriapoda (Chilopoda and Diplopoda). Classes are separated by body covering, number and type of limbs, respiratory organs and mode of reproduction.
Skills	Observing and grouping animals using distinguishing characteristics; completing classification tables; constructing and using a simple dichotomous key; constructing explanations with Claim-Evidence-Reasoning; arguing from evidence; communicating clearly to peers. Core competencies: Communication and Collaboration (speaking clearly to peers while discussing distinguishing characteristics of specimens); Creativity and Imagination.
Attitudes	e) appreciate the diversity of plants and animals in nature. Values: Respect (considerate of others' opinions when groups disagree on classifications). Responsibility (cares for the environment and the animals handled). PCI: Animal Welfare (handles animals with care while observing and grouping them into Phylum Chordata and Arthropoda using distinguishing characteristics).
Key Inquiry Question	1. How are plants and animals classified?
Purpose in Storyline	In Lessons 1-5 learners met the problem that their own groupings (size, colour, habitat, usefulness as food) disagreed, and they learned about kingdoms, plant divisions and animal phyla. In Lesson 6 they see that phyla are still broad: a goat, chicken, lizard and the school's fish pictures are all Chordata, and a mosquito and butterfly are both Arthropoda, yet they differ. Classes are the next level. This raises the question for Lesson 7: 'How can we identify an unknown animal accurately?', which leads to dichotomous keys.
Safety Notes	Use only locally available, harmless specimens (preserved or dead insects, snail shells, earthworms from the school garden, feathers, fish bones or pictures). Do not collect dangerous or unknown organisms (snakes, scorpions, wasps, bees, centipedes with bites). Do not handle live mosquitoes. Handle any live animal gently with a spoon or gloved hands, return it to where it was found, and wash hands with soap and water afterwards. Where specimens are unavailable, use drawings, photographs or textbook illustrations. Learners may discuss in Kiswahili or mother tongue, but record scientific terms in English.

B. LESSON OVERVIEW

Anchoring phenomenon: the school collection (maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat) shows living things that cannot be grouped in one way that works for everyone. Today learners return to the animals in that collection. They predict how a goat, chicken and lizard might be grouped together, and how a mosquito and butterfly relate. Then, in a Jigsaw, expert groups examine specimens, pictures and charts for one class group (fish, amphibians, reptiles, birds, mammals, and arthropod classes). They compare body covering, limbs, respiration and reproduction, and complete a classification table. Home groups then teach each other, test their ideas with a dichotomous-key challenge on the phenomenon animals, and defend a classification using CER. The lesson closes by updating the Driving Question Board and the class model of classification with the new idea that phyla are divided into classes using more specific characteristics. Materials: animal pictures/specimens, charts, classification cards, video (optional, offline copy if available), textbooks shared by groups, manila paper, markers.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy class chordata arthropoda  HTML: Classifying Classes (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners recall the school collection from the phenomenon. Display or draw the animals: earthworm, snail, mosquito, butterfly, chicken, lizard and goat. Individually, each learner writes a prediction in their notebook: 'The chicken, lizard and goat are all in Phylum Chordata. Which features could we use to split them into smaller groups? Which feature would separate the mosquito from the butterfly, if any?' They draw a quick sketch grouping the 7 animals into smaller groups and give a reason for each group. Then Think-Pair-Share: pairs compare and note one place they disagree. Kiswahili may be used in discussion (e.g. 'ndege', 'mjusi', 'mamalia'), with scientific terms recorded in English.	Display the animals or the cards at the front. Say: 'Last lesson we placed animals in phyla. A goat, a chicken and a lizard are all Chordata. Are they the same? Think quietly and write your prediction first.' Give WAIT TIME of 10-15 seconds after each question before anyone speaks. Allow 3 minutes for individual writing, then 3 minutes for pairs. Cold-call 4 pairs (not volunteers): 'What feature did you use to split the chicken from the goat?' Record all predictions on the board without judging them as right or wrong. Ask: 'Whose grouping is different from your partner's? Why?' Circle features that appear repeatedly (skin covering, legs, wings, how it breathes, eggs or live young).	Predict-Observe-Explain (POE) combined with Think-Pair-Share. Making predictions first exposes learners' own grouping ideas (size, colour, habitat, usefulness as food) and creates a need for shared, observable characteristics. Think-Pair-Share gives every learner time to think and rehearse before speaking.	Look for: (1) at least 80% of learners write a prediction with a reason before talking; (2) number of pairs who name a structural feature (covering, limbs, breathing, reproduction) versus non-structural ones (size, colour, food use); (3) pairs who notice that their groupings differ and say so. Note misconceptions, such as 'all animals that fly are birds' (butterfly) or 'worms and snakes are the same' and revisit them in the Explain Phase.
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: taxonomy class chordata arthropoda  HTML: Classifying Classes (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar	Jigsaw, Part 1 (expert groups, 20 minutes). Class is divided into 6 expert groups of 5-6 learners. Each expert group receives a specimen/picture set, a chart and classification cards for ONE group: (1) Pisces (tilapia or omena from Lake Victoria, pictures of Nile perch); (2) Amphibia (frog/toad pictures); (3) Reptilia (lizard/agama pictures, tortoise, snake pictures only); (4) Aves (chicken feathers, egg, pictures of flamingo at Lake Nakuru); (5) Mammalia (goat or cow hair, skull/picture, pictures of a lion from Maasai Mara); (6) Arthropoda classes (dead mosquito and butterfly, spider picture, crab picture from Mombasa, millipede/centipede pictures).	Distribute materials and say: 'You are now the experts. Your task: find the features that make your group different from every other group. Look, touch gently, and record what you SEE, not what you guess.' Circulate with prompts: 'Show me where it breathes. What does its skin feel like? How many legs? Does it lay eggs or give birth?' Use WAIT TIME of 10-15 seconds after each prompt. Prompt groups that are stuck: 'What does a mosquito have that a spider does not?' (antennae, 3 pairs of legs vs 4 pairs). For home groups, say: 'Teach, do not read. Each expert has 2 minutes. Others must ask one question.' Cold-call 3 home groups to share one feature that separates two classes. Stop at	Jigsaw cooperative learning. Each learner becomes responsible for one piece of evidence, so everyone must contribute to the complete classification table. Comparing the same four characteristics across all classes helps learners see patterns (e.g. feathers only in birds, hair only in mammals, three body parts and six legs in insects) instead of memorising lists.	Check expert tables for accuracy: correct covering (scales, moist skin, feathers, hair, exoskeleton), limbs, respiration (gills, lungs, skin, tracheae, book lungs) and reproduction. Observable indicators: learners point to the actual feature on the specimen when explaining; learners use class names correctly; every home group member speaks at least once. Look for the error 'frogs have scales' or 'insects have 8 legs' and correct it by asking learners to count or recheck.

readings	Groups examine the specimens carefully and complete a classification table with columns: body coverings, limbs/appendages, respiration organs, reproduction (eggs or live young; external or internal fertilisation), and one other feature. Learners handle animals gently and return live animals (such as earthworms and snails) to the garden. If a video is available offline, groups may watch a short clip to check breathing and reproduction details; otherwise they use textbook pages. Jigsaw Part 2 (home groups, 15 minutes): learners regroup in mixed home groups with one expert from each class. Each expert teaches their class's features in 2 minutes while others fill in a master table.	the 10-minute mark to check that tables are filling in. Remind learners about Animal Welfare: 'Be gentle. These are living things.'		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: taxonomy class chordata arthropoda  HTML: Classifying Classes (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners use their master table to build a claim about the animals in the phenomenon. Step 1 (Dichotomous-key challenge, 10 minutes): each home group writes a simple dichotomous key with 3-4 questions that separates the phenomenon animals (mosquito, butterfly, earthworm, snail, chicken, lizard, goat) into classes, using paired statements such as '1a. Has feathers - Aves; 1b. No feathers - go to 2'. Groups swap keys and test each other's keys with the classification cards. Step 2 (CER, 8 minutes): each learner writes a Claim-Evidence-Reasoning response to: 'Is a bat a bird or a mammal?' or 'Is a mosquito an insect or an arachnid?' Claim: the class of the animal. Evidence: at least two observed features from the tables. Reasoning: why those features define the class. Step 3: groups defend their classification when	Say: 'Your key must work for someone who has never seen these animals. Only use features you can observe.' Give WAIT TIME of 10-15 seconds after the question 'Why can't we use colour in our key?' Before key swapping say: 'If the other group gets stuck, the problem is in the key, not the group.' During CER circulate and say: 'Show me the evidence in your table. Which feature proves your claim?' Cold-call 3 learners to read a CER aloud, then ask 2 other learners: 'Do you agree? What evidence would change your mind?' Introduce correct terms at this point (not before): Pisces, Amphibia, Reptilia, Aves, Mammalia; Insecta, Arachnida, Crustacea, Myriapoda. Clarify tricky cases: a bat is a mammal (hair, milk, live young); a snail and earthworm are not arthropods or chordates (they belong to other phyla); a tilapia breathes with gills.	Dichotomous-key challenge plus Claim-Evidence-Reasoning (CER). Writing a key forces learners to choose clear, observable, either/or characteristics, which shows why scientists use shared features. CER trains learners to defend a classification with evidence and to revise their ideas when other groups present better evidence.	Use a 3-level CER check: Level 1 states a claim only; Level 2 gives a claim with a correct feature; Level 3 links two or more features to the class definition with reasoning. Aim for most learners at Level 2 or 3. Observe whether keys use observable paired statements and whether another group can use them to reach the correct class. Note which learners revise their ideas after challenge. Exit check: 'Name one feature that separates reptiles from amphibians.'

	challenged by another group, and revise if the evidence supports a change. Learners link back: 'Why did our first groupings by size or colour disagree, and why do the class features not disagree?'	Model the correct sentence frame: 'I claim that ____ belongs to class ____ because ____, which shows ____.' Use Kenyan examples (omona, tilapia, chameleon, ostrich, flamingo, goat, grasshopper, crab).		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: taxonomy class chordata arthropoda  HTML: Classifying Classes (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board. Individually they write on sticky notes (a) one new thing they learned about classes of animals, and (b) one new question that has appeared (e.g. 'What if an animal has features of two classes?', 'How do we identify an animal we have never seen?', 'Where do viruses fit if they are not animals or plants?'). In groups they place the notes on the board under headings such as 'Answered', 'Partly answered' and 'New questions'. The class discusses how this lesson answers part of 'How are plants and animals classified?' and leads to the next lesson question: 'How can we identify an unknown animal accurately?'	Say: 'Look at our board from Lesson 1. Which questions can we now answer? Take 30 seconds of silent thinking.' Allow WAIT TIME of 10-15 seconds. Cold-call 4 learners to move a question from 'Unanswered' to 'Partly answered' or 'Answered' and justify it with evidence from today. Ask: 'What new question did the classes raise?' Highlight the question 'How can we identify an unknown animal accurately?' and say: 'In Lesson 7 we will use dichotomous keys more carefully to answer it.' Ensure that Respect is shown by thanking learners for questions that differ from yours.	Driving Question Board. Tracking questions and answers makes learning visible and shows learners how today's evidence connects to the unit driving question, keeping the lesson phenomenon-driven and learner-led.	Check that sticky notes link to evidence from today (e.g. 'birds have feathers and lay eggs, so...'), that at least one new question per group goes beyond recall, and that learners can justify moving a question's status. Record which questions show lingering misconceptions.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: taxonomy class chordata arthropoda  HTML: Classifying	Learners revise the class classification model started in earlier lessons. On manila paper, each group updates a tree/flow diagram that began with the phenomenon animals: Kingdom Animalia, then phyla (Chordata, Arthropoda, others), and now classes with the key identifying features beneath each class (body covering, limbs, respiration, reproduction). They place the phenomenon animals (goat, chicken, lizard, mosquito, butterfly, earthworm, snail) on the correct	Say: 'Go back to our first puzzle: how can any person anywhere group these living things the same way? Show on your model how classes help.' Give WAIT TIME of 10-15 seconds after this question. Circulate and ask: 'Which feature put the lizard in Reptilia and not Amphibia?' Use the gallery walk: groups rotate every 90 seconds; instruct them 'Leave one star (agreement based on evidence) and one question.' Cold-call 3 groups to explain one change they made to their model from last	Model revision with a mini gallery walk. Revising a visible model shows how new evidence changes ideas, and peer feedback helps learners spot gaps and strengthen their explanations of how scientists organise living things.	Look for: animals placed on correct class branches; at least two correct features per class; model revisions linked to evidence; gaps identified honestly (e.g. 'we cannot yet identify unknown animals'). Quick check: ask 3 learners to place the bat, the crab and the tilapia on the model and justify each. Use the summary table to confirm understanding.

Classes (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	branches and mark with a different colour the ones that are not Chordata or Arthropoda. They add a short note: 'What this model explains' and 'What it cannot yet explain' (e.g. how to identify an unknown animal quickly). Groups then do a mini gallery walk (5 minutes), leaving one comment of agreement and one question on each other's models. Finally, learners complete the summary table prompt in their notebooks.	lesson and the evidence that caused it. Close with: 'Phyla are broad groups. Classes use more specific characteristics. But a model is not complete until it helps us identify an animal we have never seen. That is tomorrow's question.' Collect models and tables for later assessment.		
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D. TEACHER REFLECTION

After the lesson reflect: Did learners use observable features rather than size, colour or habitat when grouping? Which classes caused the most confusion (e.g. amphibians versus reptiles, insects versus arachnids)? Were all learners active in the Jigsaw, and did specimens or pictures provide enough evidence for each group? Which questions or WAIT TIME moments produced the deepest thinking? How well did the groups' CER responses connect features to class definitions? Did animals stay safe and were they returned to the environment? What adjustments are needed before Lesson 7, such as more practice with paired statements for dichotomous keys, additional local specimens (grasshopper, tilapia, chameleon) or additional use of Kiswahili for early discussion?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a goat, chicken, lizard and fish (all Chordata) differ in body covering (hair, feathers, scales), limbs, respiration (lungs, gills) and reproduction (eggs or live young), and that a mosquito, butterfly and crab (all Arthropoda) differ in number of legs, body parts and antennae.
What did I learn?	We learned that phyla can be divided further into classes using more specific characteristics: Chordata into Pisces, Amphibia, Reptilia, Aves and Mammalia; Arthropoda into Insecta, Arachnida, Crustacea and Myriapoda.
How does this explain the phenomenon?	We can explain that the school animals can be organised into classes using shared, observable features (not size, colour or habitat), so anyone can classify them the same way, but we still need a reliable method to identify an unknown animal accurately, which leads to the next lesson.

LESSON 7 (80 minutes (double lesson)): Dichotomous keys for animals and viruses

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners construct and use a dichotomous key to identify the school-environment animals from the anchoring phenomenon (mosquito, butterfly, earthworm, snail, chicken, lizard, goat), and brainstorm the general characteristics and economic importance of viruses, so they can revise their ideas about how to identify and classify living organisms accurately.
Knowledge	By the end of the lesson the learner should be able to: (a) outline the principles of constructing a dichotomous key (paired contrasting statements, observable features, one choice per step); (b) use a key to identify animals and place them in Phylum Chordata or Arthropoda; (c) state the general characteristics of viruses (non-cellular, only reproduce inside host cells, protein coat around DNA or RNA, no cell division or metabolism outside a host) and relate viruses to their economic importance in health, agriculture and biotechnology.
Skills	Constructing explanations and designing solutions (NGSS SEP 6): building a key. Analysing and interpreting data (SEP 4): comparing group classifications. Engaging in argument from evidence (SEP 7): CER about a virus. Developing and using models (SEP 2): revising the classification model. Communication and Collaboration: speaks clearly to peers while discussing distinguishing characteristics of various specimens.
Attitudes	Respect: considerate of others' opinions while brainstorming on the characteristics of viruses and their economic importance. Animal Welfare (PCI): handles animals with care while observing and grouping them into Phylum Chordata and Arthropoda. Appreciates the diversity of plants and animals in nature.
Key Inquiry Question	How are plants and animals classified? How are viruses important in nature? (Driving question: How can we identify and classify living organisms accurately?)
Purpose in Storyline	Lessons 1-6 showed that learner-made groupings by size, colour or usefulness differ between groups, and that shared structural features (plants, Chordata, Arthropoda) give a standard basis for grouping. Lesson 7 shows how a dichotomous key turns those features into a tool that lets anyone anywhere reach the same identification, and extends the system to viruses, which challenge the idea of a living organism. It sets up Lesson 8: How can our classification system help us explain biodiversity and the importance of viruses?
Safety Notes	Use animal cards, photographs and preserved or already-dead specimens wherever possible. Do not collect dangerous or unknown organisms (snakes, scorpions, bees, wasps, unknown caterpillars). Handle any live earthworm, snail or insect gently, return it to where it was found, and wash hands with soap and water afterwards. Do not handle the mushroom or any unknown fungus. Do not touch sick or dead animals. Virus work uses diagrams only; no cultures are used.

B. LESSON OVERVIEW

Learners return to the phenomenon: 'If all these organisms are living things, how can scientists organise them into groups so that any person anywhere can identify and study them?' They first predict how they would identify an unfamiliar animal from only a written description. Using animal cards (mosquito, butterfly, earthworm, snail, chicken, lizard, goat, plus a Kenyan extra such as a tilapia from Lake Victoria and a grasshopper), groups build their own dichotomous key, swap keys with another group, and test whether a stranger reaches the same answer, which directly confronts the problem that subjective groupings (size, colour, usefulness) fail. They then compare their key with a prepared key and use it to place animals in Chordata or Arthropoda. In the second half, learners examine virus diagrams, readings and a short video (if available; a printed reading is the offline alternative) to build a list of virus characteristics and economic importance (maize lethal necrosis disease in Kenyan maize, cassava mosaic disease, HIV, measles and polio vaccines, COVID-19, and the use of viruses in biotechnology) and write a CER on whether a virus is a living organism. The Driving Question Board is updated and the class model of classification is revised to include keys and viruses.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key couplet	Display the cards of a mosquito, butterfly, earthworm, snail, chicken, lizard and goat, and ask: 'Imagine a visitor from Kisumu who has never seen your local organisms. You can only send them a written message. How can you make sure they name the same animal as you?' Learners individually write a prediction in their notebooks: (1) what kind of statements would work best in the message (e.g. 'it is big' versus 'it has six legs'), and (2) whether a virus can be placed in the same classification as the goat or the maize plant. They then think-pair-share and may use Kiswahili for the initial discussion. Scientific terms are recorded in English.	Say: 'In Lesson 1, groups made different groupings for the same organisms. Today we find out how to make a tool that gives the same answer for everyone.' Pose the question, then WAIT TIME of 12 seconds in silence before any talk. Give 2 minutes for individual writing, then 2 minutes for pairs. Cold-call 4 pairs: 'What kind of statement did you choose, and why?' Write phrases like 'big', 'beautiful', 'useful' and 'six legs', 'has a backbone' on the board in two columns without labelling them yet. Ask: 'Which column would two different people read the same way?' Do not tell the answer; record predictions for revisiting at the end of the lesson. Ask: 'Is a virus in the same group as the goat? Raise your hand for yes, no, not sure.' Record the count.	Predict-Observe-Explain with Think-Pair-Share: committing to a prediction in writing before instruction surfaces the idea that vague descriptions (big, beautiful) differ between people, and makes learners notice the conflict when their predictions are tested against evidence later.	Look for: (1) learners who write subjective features (size, colour, usefulness as food) versus those who write observable, countable features (number of legs, wings, backbone, shell); (2) the tally of yes/no/not sure on viruses as a baseline; (3) pairs who disagree and can say why. Note the names of 5 learners with subjective-only predictions for support during the construction task.
Observe Phase  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key	Part A (about 30 minutes): Groups of 4 receive 8 animal cards (mosquito, butterfly, grasshopper, earthworm, snail, chicken, lizard, tilapia or goat) with photos and short observation notes (legs, wings, backbone, body segments, shell, body covering). Each group (1) lists observable differences, (2) builds a dichotomous key with paired contrasting statements, each leading to another couplet or an animal name, for example 'Backbone present / Backbone absent', then 'Feathers present / Scales or hair present', 'Six jointed legs / No jointed legs', (3) swaps keys with another group and tests	Circulate with a checklist. Before starting say: 'A good key is like a path with forks. At every fork the visitor has only two choices, and both must be things anyone can see or count.' Model one couplet on the board with the earthworm and the snail. Prompt struggling groups with: 'Could two people disagree on this statement? If yes, change it.' When a group uses 'small / big', say: 'Small compared to what? Show me a feature that I can count.' Give WAIT TIME of 10 seconds after each question before offering hints. Set the timer: 10 minutes to build, 8 minutes to swap and test, 7 minutes to revise.	Dichotomous-key challenge: converting shared, observable features into a fork-by-fork tool shows learners that classification works when statements are contrasting and objective, directly solving the problem from the phenomenon that subjective groupings differ between groups. Swapping keys is a built-in test of reliability. For viruses, comparing diagrams with cell structures builds understanding through contrast.	Observable indicators: (1) each couplet has two clearly contrasting statements using a countable or visible feature; (2) the swapped group reaches the intended animal for at least 3 of 3 mystery cards; (3) learners can correct a flawed couplet after testing; (4) animals are correctly placed in Chordata (chicken, lizard, goat, tilapia) or Arthropoda (mosquito, butterfly, grasshopper), with earthworm and snail correctly noted as neither; (5) in Part B, learners list at least 3 virus characteristics and 3 importance examples with at least one beneficial use. Record each group's tally of correct

couplet	it by identifying 3 'mystery' animals, and (4) records where the key failed or was ambiguous and fixes it. Groups then compare with a prepared key and sort the animals into Chordata or Arthropoda. If the school compound has safe specimens (e.g. a dead insect, a snail shell), learners may use them in addition. Part B (about 25 minutes): Learners use virus diagrams (e.g. bacteriophage, HIV, tobacco mosaic virus), textbook readings (groups share textbooks), and a short video if available, to build a table of virus characteristics (size, protein coat, genetic material, cannot reproduce outside a host cell) and a list of their importance: harmful (maize lethal necrosis disease and cassava mosaic disease in Kenyan farms, measles, polio, HIV, COVID-19, Newcastle disease in chicken, foot and mouth disease in cattle) and useful (vaccines, gene transfer in biotechnology, bacteriophages used against bacteria, biological control of pests). Learners should be considerate of others' opinions as they brainstorm.	Ask each group to nominate a tester who is not the author of the key. Remind learners to handle all specimens gently and return any live animals to where they were found (Animal Welfare). In Part B, hand out one virus diagram per group and ask: 'Where is the nucleus? Where are the mitochondria? What do you see instead?' Cold-call 5 groups to share one characteristic each and ask whether any group disagrees. Where internet is unavailable, use the printed virus reading and teacher-drawn diagram on the chalkboard. Ensure respectful turn-taking with the phrase: 'Build on the idea before you challenge it.'		identifications on your checklist.
Explain Phase  VIDEO: 25 Dichotomous Key I <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this</i>	Learners individually write a Claim-Evidence-Reasoning (CER) statement for the question: 'Is a virus a living organism?' Claim: yes/no or 'it depends'. Evidence: at least two observations from the diagrams and reading (e.g. no cell structure, reproduces only inside a host cell, has genetic material). Reasoning: link to the characteristics of living things used in earlier lessons (nutrition, respiration, growth, reproduction, excretion, response, movement). They then think-pair-share, and 3	Say: 'Make a claim you can defend with evidence from today. Opinions without evidence will not count.' Provide a CER frame on the board: 'I claim ... because ... which shows ... so ...'. WAIT TIME of 15 seconds after posing the CER question before writing begins. Give 5 minutes for writing, 3 for pairs. Cold-call 3 learners with different claims and ask the class: 'Who can add a piece of evidence or challenge the reasoning politely?' Probe: 'You said viruses are not alive. Which	Claim-Evidence-Reasoning (CER) with Think-Pair-Share: requiring learners to link specific evidence to a general rule of what counts as living moves them from opinion to scientific argument, and public critique with respectful disagreement builds shared understanding. Returning to the Predict phase prediction helps learners see how their thinking has changed.	Use a 3-point rubric: 1 = claim only; 2 = claim with one piece of evidence; 3 = claim, at least two pieces of evidence and reasoning linked to characteristics of living things. Aim for at least 70% of the class at level 2 or above; note learners at level 1 for follow-up. Listen for correct use of the terms couplet, host cell, protein coat, and for the ability to say why a key is more reliable than a subjective grouping. Collect 5 CERs as exit evidence.

activity.  Search ARES for readings Search terms: taxonomy dichotomous key couplet	volunteers present to the class. Next, learners explain in pairs why the key they built works better than their earlier groupings by size, colour or usefulness, citing the evidence from their key test. Finally, they discuss in groups: 'How would a farmer in Trans Nzoia use a key and knowledge of viruses to protect a maize crop?'	characteristic of living things does it fail, and which does it appear to show only inside a host?' Probe the key: 'Why did the swapped key give the same answer for everybody, when our earlier groupings did not?' Confirm the target idea: a key uses paired contrasting observable features; viruses are non-cellular and depend on a host cell, so they are not placed in the five kingdoms in the same way as organisms. Link to the economic examples: maize lethal necrosis, cassava mosaic, vaccines. Correct misconceptions: 'viruses are tiny bacteria' and 'antibiotics kill viruses'.		
Driving Question Board (DQB) Creation  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy dichotomous key couplet	Learners return to the Driving Question Board. In their groups, they (1) place a sticky note answering part of the driving question ('How can we identify and classify living organisms accurately?') with new evidence from today, such as 'A dichotomous key uses paired, observable contrasting features'; (2) mark earlier questions that are now answered (for example 'Why do our groupings differ?'); and (3) add at least one new question, such as 'How does classification help us explain how many different organisms live in Kenya?', 'Why do viruses keep changing (e.g. COVID-19 variants)?', or 'Can a key be made for plants from our school?'. Each group contributes at least one note.	Say: 'Look at our board. Which questions can we now answer with evidence? Which new ones does today raise?' Give 2 minutes of silent thinking, then WAIT TIME of 10 seconds before taking contributions. Cold-call 4 groups to read and place a note and explain the evidence behind it in one sentence. Move answered questions to an 'Answered' column with a coloured marker and ask: 'What was the evidence?' Push for questions that connect to the phenomenon: 'Which new question helps us explain why Kenya has so many different kinds of organisms and why viruses matter to farmers and health workers?' Highlight the question that leads to Lesson 8: 'How can our classification system help us explain biodiversity and the importance of viruses?' If the DQB is unavailable, use a chart paper or a chalkboard section.	Driving Question Board: a visible, class-owned record of questions and answers connects each lesson's evidence to the unit's driving question and gives learners ownership of what to investigate next. Linking notes to evidence builds the habit of answering questions with data rather than recall.	Check that notes (1) cite evidence from today (a key test result or a virus characteristic); (2) show progress from earlier questions, such as moving from 'which grouping is best?' to 'which features are best for a key?'; (3) include at least one new question that is testable using the local environment or reading. Note groups that only copy facts from the textbook rather than posing questions.
Model Building	Learners revise the class and individual models (a concept map	Say: 'Use a different colour to show what has changed in your	Model revision: updating a visible model with a different colour	Check the model for: (1) a correct example of a dichotomous key

Phase  VIDEO: 26 Dichotomous Key li Source: Seavuria  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: taxonomy, dichotomous key, couplet	or flow diagram started in Lesson 1 and revised in later lessons) of how living organisms are identified and classified. Each learner (1) adds a dichotomous key branch showing how to identify animals from the phenomenon collection (maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) at the level of Chordata and Arthropoda; (2) adds a box for viruses showing that they are non-cellular, need a host, and are important in health, agriculture and biotechnology; and (3) uses a different colour pen to mark what they changed since the Predict phase and why. They then revisit their opening prediction and write two sentences: 'I used to think ... Now I think ... because ...'.	thinking. Scientists keep track of how their ideas change.' Give 10 minutes of individual work, and circulate to ask 3 learners: 'What evidence made you add this part of your model?' WAIT TIME of 10 seconds after each question. Then pair learners for 3 minutes to peer-review each other's models using the prompts: 'Is every link supported by evidence? Is anything missing?' Cold-call 3 learners to share one change. Revisit the Predict phase tally on whether viruses belong with the goat and ask learners to revise their vote with the evidence. Finish by posing the next question: 'Our system can name and group organisms. How might it help us explain the huge variety of life, and the importance of viruses, in Kenya?' Collect models for review.	makes the change in thinking explicit, and requiring evidence for each addition builds the habit of revising explanations in light of data. Peer review adds an external check on reasoning.	linked to observable features; (2) viruses represented as non-cellular and host-dependent, with at least two examples of importance (one harmful, one beneficial); (3) a before/after statement that cites evidence from today; (4) linkages to the original phenomenon organisms. Use a quick 3-level rubric (missing / partial / complete) and record how many learners are at 'complete' to plan Lesson 8.
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D. TEACHER REFLECTION

After the lesson reflect: (1) Did learners choose observable, countable features for the key, or did some stay with subjective ones like 'big' or 'beautiful'? Which prompts moved them? (2) Did the swap-and-test step reveal ambiguity in couplets, and did groups fix it without being told? (3) How did the class vote on whether a virus is alive change between Predict and Model Building? (4) Were any learners unable to take part because of limited textbooks or a lack of video access, and did the printed reading and chalkboard diagram work as alternatives? (5) Were animals handled with care and returned safely? (6) Which Kenyan examples (maize lethal necrosis, cassava mosaic disease, vaccines) engaged learners most, and what should be added in Lesson 8 on biodiversity? Note learners who need more support with writing couplets or CER before the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? (Record how your dichotomous key worked when another group used it, which couplets were clear or confusing, which animals fell in Chordata and Arthropoda, and what the virus diagrams showed about structure.)
What did I learn?	What did we learn? (A dichotomous key uses paired contrasting observable features to identify organisms reliably; viruses are non-cellular, have a protein coat and genetic material, and reproduce only inside host cells; viruses are important in health, agriculture and biotechnology.)
How does this explain the phenomenon?	How does this help explain the phenomenon? (Explain why a key gives the same answer for everyone while groupings by size, colour or usefulness do not, and how knowing about viruses shows that classification must be based on shared observable

	characteristics. Raise a question for Lesson 8 on biodiversity.)
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LESSON 8 (80 minutes (2 x 40 min lessons)): Synthesis and explanation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise evidence from Lessons 1-7 to explain how living organisms can be identified and classified accurately, and compare their final model with their Lesson 1 model.
Knowledge	By the end of the lesson the learner should be able to: a) outline the general characteristics of organisms in Divisions Bryophyta, Pteridophyta and Spermatophyta (including Gymnospermae and Angiospermae); b) classify organisms into phyla and classes of Kingdom Animalia (Chordata and Arthropoda); c) construct and use a dichotomous key to identify organisms; d) relate viruses to their economic importance in health, agriculture and biotechnology.
Skills	Developing and using models (revising the classification model); Constructing explanations using Claim-Evidence-Reasoning; Engaging in argument from evidence; Obtaining, evaluating and communicating information; Collaboration and communication when presenting keys.
Attitudes	Appreciate the diversity of plants and animals in nature. Respect: be considerate of others' opinions while discussing and critiquing group models. Responsibility: care for the environment and the specimens used. Core competencies: Communication and Collaboration (speak clearly to peers while discussing distinguishing characteristics of specimens); Creativity and Imagination (experiment with varied ideas when designing keys and models). PCIs: Environmental Conservation (conserve biodiversity by collecting organisms responsibly); Animal Welfare (handle animals with care).
Key Inquiry Question	1. How are plants and animals classified? 2. How are viruses important in nature? Driving question: How can we identify and classify living organisms accurately?
Purpose in Storyline	This is the final lesson. Learners return to the school-environment phenomenon (maize, bean, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) that first puzzled them because every group classified differently. They now use evidence from all lessons to show that a standard system based on shared observable characteristics lets anyone anywhere identify and study organisms. It closes the DQB and leads to the next question: How can classification knowledge be applied outside the classroom?
Safety Notes	Use only familiar, safe, locally available specimens or cards/photographs; do not collect dangerous or unknown organisms (e.g. snakes, scorpions, unknown fungi). Wash hands after handling any specimen, soil or plants. Do not taste mushrooms or any specimen. Handle live animals gently and return them to where they were found (Animal Welfare). Collect no more plant material than needed and never uproot rare plants (Environmental Conservation). Keep walkways clear during the gallery walk.

B. LESSON OVERVIEW

Lesson 8 is the culminating synthesis lesson. Learners begin by predicting how they would now classify the original collection of organisms, then use a final classification challenge (organism cards/specimens plus a dichotomous key) as evidence of how far their thinking has moved. In the Explain phase they write and defend a Claim-Evidence-Reasoning Final Explanation answering the driving question 'How can we identify and classify living organisms accurately?' The class then completes the Driving Question Board, marking which questions are answered and which new ones remain, and finally compares their Lesson 1 model with the final model. The lesson ends with the new question: How can classification knowledge be applied outside the classroom? Learners may discuss in Kiswahili or mother tongue, but scientific terms are recorded in English. The lesson does not depend on internet; the video is optional and can be replaced by charts and learners' notes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview Source: CK-12  Search ARES for similar videos  HTML: Linnaean Classification (Flexbook) Source: CK-12  Search ARES for similar readings	Learners view the display of organisms from the school environment (maize plant, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) as real specimens, photographs or cards. In pairs they individually write a prediction: 'Mixed in with a new set of 6 unfamiliar organisms from Kericho, Kisumu or Mombasa, could a learner from another school classify them exactly as we do? What would we need so that the answer is yes?' They then share with a partner (Think-Pair-Share), discussing first in Kiswahili or mother tongue if helpful, and record key terms in English.	Display the organisms and say: 'In Lesson 1 we asked: if all these are living things, how can scientists organise them so anyone, anywhere, can identify them? Today you will answer that with evidence.' Pose the prediction question and give WAIT TIME of 10-15 seconds of silent thinking, then 2 minutes to write. Pair learners for 3 minutes. Cold-call 4 pairs: 'What would another learner in Kisumu need to get the same groups as you?' Listen for 'same characteristics', 'a key', 'standard names'. Record answers on the board without evaluating. Say: 'Let us test these predictions today.'	Predict-Observe-Explain (POE) reprise with Think-Pair-Share: Writing predictions before the challenge makes learners commit to ideas drawn from earlier lessons, so they can later see how evidence confirms or changes their thinking. Pairing lets learners rehearse scientific vocabulary before whole-class talk.	Check that at least 80% of predictions mention a shared, observable characteristic or a key (not size, colour or usefulness as food). Indicators: learners use terms like 'dichotomous key', 'division', 'phylum', 'class'; note learners who still propose arbitrary groupings for follow-up. Tally the ideas on the board for comparison at the end.
Observe Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Linnaean Classification Practice Source: CK-12  Search ARES for similar readings	Final classification challenge (groups of 4). Each group receives organism cards/specimens (maize, bean, fern, moss, pine/cypress leaf or cone from Kenyan highlands, mushroom picture, mosquito, butterfly, earthworm, snail, chicken, lizard, goat, plus 2 new unfamiliar cards, e.g. a sisal plant and a tilapia from Lake Victoria). Task 1: sort them into Kingdom Plantae and Kingdom Animalia and then into Divisions (Bryophyta, Pteridophyta, Spermatophyta; Gymnospermae/Angiospermae; monocot/dicot) and Phyla/Classes (Chordata: fish, amphibians, reptiles, birds, mammals; Arthropoda: insects, arachnids, crustaceans, myriapods). Task 2: build or complete a dichotomous key for at least 6 organisms and exchange it with another group,	Say: 'Your challenge is to classify the same organisms as before, but this time using shared, observable characteristics.' Distribute cards and rubrics. Circulate and ask: 'Which characteristic did you use for that split? Could a learner in Mombasa observe it too?' and 'Where does the mushroom fit, and why does it not fit Plantae?' WAIT TIME 10-15 seconds after each question. Cold-call at least 6 learners across groups, including quieter learners. Prompt for Animal Welfare: 'How are we handling the live earthworm or snail?' Remind groups to use only safe specimens and to return live ones to the school garden. During key exchange say: 'The test of a good key is whether a stranger can use it. Note exactly where they got stuck.' Move through the stations	Dichotomous-key challenge with gallery presentations: Using and testing another group's key shows that accurate identification depends on clear, observable, yes/no characteristics, directly addressing the Lesson 1 problem where each group's classification differed. Gallery presentation exposes learners to multiple models so they can compare evidence.	Observable indicators: (1) groups correctly place at least 10 of 13 organisms into the right division, phylum or class; (2) keys use observable paired statements (e.g. 'has wings / no wings', 'seeds in fruit / seeds not in fruit', 'one cotyledon / two cotyledons', 'vertebral column present / absent'); (3) the exchanging group identifies the mystery organism correctly; (4) learners can name one importance of arthropods (pollination, pests, disease vectors such as mosquitoes) and one of viruses. Record on a checklist and note misconceptions, e.g. 'mushroom is a plant' or 'viruses are cells'.

	who must use it to identify an unlabelled 'mystery' organism. Task 3: record where the key was unclear. Groups present their models/keys in a short gallery-style rotation (2 minutes per station), with learners recording evidence in their notes and tick-checking the rubric. Groups also add a one-line note on viruses: why they are not placed in either kingdom, and their importance (e.g. vaccines, maize streak virus in crops, biotechnology). Optional 5-minute video or chart review to confirm characteristics.	asking each group to justify one choice. For viruses ask: 'Give one harm and one benefit of viruses to Kenyans.' Listen for maize streak virus, HIV/measles/COVID-19, vaccines, gene therapy and biotechnology.		
Explain Phase  VIDEO: 01 Overview Of Kingdom Plantae <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Animal Classification (Flexbook) Source: CK-12  Search ARES for similar readings	Learners write an individual Final Explanation answering the driving question using Claim-Evidence-Reasoning (CER). Claim: a statement answering 'How can we identify and classify living organisms accurately?' Evidence: at least three pieces from the lessons and the challenge (e.g. moss lacks vascular tissue while fern has vascular tissue and spores; maize has parallel veins and one cotyledon whereas bean has net veins and two cotyledons; mosquito has 3 pairs of legs and a jointed exoskeleton so is in Arthropoda, while chicken has a backbone so is in Chordata; a key gave the same answer for two different groups). Reasoning: explains why shared, observable characteristics and standard names produce the same grouping for any learner, unlike groupings by size, colour or food use. They also include one sentence on the importance of viruses. Then they do Think-Pair-Share to critique a partner's CER using the rubric and revise.	Display the CER frame: 'I claim that ... The evidence is ... This shows that ... because ...'. Say: 'Your claim must answer the driving question, and your evidence must come from what we observed, not from memory only.' Give 8 minutes of silent writing. WAIT TIME 10-15 seconds when asking: 'Why did our groups disagree in Lesson 1 but agree today?' Cold-call 3 learners to share claims and 3 to share the strongest evidence. Probe: 'Is size or colour good evidence for classification? Why not?' and 'How does a dichotomous key help someone who has never seen this organism?' Press for reasoning that links characteristics to a standard system. Use the rubric to give feedback: 'Your evidence is clear; now explain why it supports your claim.' Allow Kiswahili for discussion but require written English terms.	Claim-Evidence-Reasoning (CER) with Think-Pair-Share: CER requires learners to connect specific evidence to a general principle, building a scientifically defensible explanation. Peer critique strengthens argumentation and clarifies gaps in reasoning.	Use a 4-level rubric: (1) Claim answers the driving question; (2) at least 3 accurate pieces of evidence from Plantae, Animalia and the key activity; (3) reasoning links evidence to the need for shared observable characteristics and a standard system; (4) correct scientific terms (division, phylum, class, dichotomous key). Look for revised wording after peer feedback. Learners who give only a list of organisms without reasoning get a follow-up prompt: 'What makes this feature a reliable way to group them?'

Driving Question Board (DQB) Creation  VIDEO: 24 Six Kingdoms Of Classification <i>Related topic: not an exact match for this lesson.</i> Source: Seavurria  Search ARES for similar videos  EXERCISE: Animal Classification Practice Source: CK-12  Search ARES for similar readings	Learners revisit the class DQB built since Lesson 1. In groups they sort the question cards into 'Answered with evidence', 'Partly answered' and 'Not yet answered'. For each answered question they attach a sticky note with the evidence and the lesson where it was found (for example, 'Why do groups classify differently?' answered by Lessons 1 and 4; 'How are viruses important?' answered in Lesson 7). They then write at least one new question that goes beyond the classroom, such as: 'How can classification knowledge be applied outside the classroom, for example in farming, health and conservation in Kenya?' and add it to the board.	Point to the DQB and say: 'Let us see how far our questions have travelled.' Give each group 5 minutes to sort cards. WAIT TIME 10-15 seconds before asking: 'Which question can we now answer with confidence, and what is your evidence?' Cold-call 4 groups to present one answered question each, and 2 groups to share a question still open. Ask: 'Which unanswered question could farmers in Kericho or conservationists at Lake Naivasha help us with?' Guide learners to the new question. Celebrate respect by thanking learners who built on a peer's idea. Do not discard unanswered questions; mark them for future study.	Driving Question Board: Revisiting the DQB makes learning progress visible, shows that learner questions drove the unit, and links the final explanation to the original phenomenon while generating a new inquiry for the next topic.	Observe whether groups can match at least 80% of the answered questions to specific evidence and lesson numbers. Indicators: learners cite activities (the key exchange, monocot/dicot comparison, arthropod classes) rather than say 'the teacher said'. Check that new questions are testable and relate to application, e.g. 'How can a key help identify crop pests in Nakuru?'
Model Building Phase  VIDEO: Plants: Classification, Reproduction, and Evolution - Overview Source: CK-12  Search ARES for similar videos  HTML: Plant Classification (Flexbook) Source: CK-12  Search ARES for similar readings	Learners retrieve their Lesson 1 models (their first grouping of the organisms) and draw or build a final model, such as a classification tree/flow chart showing Kingdom Plantae and Animalia, divisions of Plantae, phyla and classes of Animalia, a short note on viruses as non-cellular, and a small dichotomous key. In pairs they compare Lesson 1 and final models using a 'Then and Now' table: What we grouped by; Why it changed; Evidence; What is still uncertain. Several learners share how their thinking changed. Finally they complete the Summary Table (What we observed / What we learned / How we can explain it).	Say: 'Take out your first model. Now draw your best model that would let any learner in Kenya reach the same groups.' Give 10 minutes. Circulate and ask: 'What did you add or change, and what evidence made you change it?' WAIT TIME 10-15 seconds. Cold-call 4 learners to describe one specific change, e.g. from grouping by 'things we eat' to grouping by 'presence of seeds'. Ask: 'Can your model place a new organism you have never seen, say a tilapia or a cypress tree?' Close by returning to the phenomenon: 'We began puzzled that our groups differed. What have we figured out?' Introduce the next question: 'How can classification knowledge be applied outside the classroom?' Set a homework thought: find one place in your home or community (farm, clinic, market in Kisumu) where classification or	Model revision: Comparing the Lesson 1 model with the final model makes conceptual change explicit, helping learners see that science improves ideas using evidence. It also consolidates that standard classification solves the original phenomenon.	Indicators of a strong final model: (1) uses shared observable characteristics rather than size, colour or use; (2) shows hierarchy from kingdom to division/phylum to class; (3) includes a functioning dichotomous key; (4) places viruses appropriately and notes their importance; (5) the 'Then and Now' table cites evidence for at least 3 changes. Learners who still use subjective groupings receive a one-to-one prompt using a specimen card.

		identification keys are used.		
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D. TEACHER REFLECTION
Which learners moved from grouping by size, colour or food use to grouping by shared observable characteristics, and what evidence in their CER or model shows it? Did the key exchange expose specific weak points (e.g. confusion between monocot and dicot features, or insect and arachnid features)? Which DQB questions remain unanswered, and how will they be used in the next lesson on applying classification outside the classroom? Were Kiswahili and mother tongue discussions supporting understanding, and did all learners record English terms? Were safety and animal welfare rules followed? What would you change about specimens or time allocation next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that organisms from our school environment (maize, bean, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard, goat) differ in structure, body covering, movement and reproduction, and that groups using size, colour or usefulness produced different groupings. When we used shared, observable characteristics and dichotomous keys, different groups reached the same results.
What did I learn?	We learned that plants are classified into Bryophyta, Pteridophyta and Spermatophyta (Gymnospermae and Angiospermae; monocots and dicots), and animals into phyla such as Chordata and Arthropoda and then classes. We learned how to build and use dichotomous keys and that viruses are non-cellular but important in health, agriculture and biotechnology.
How does this explain the phenomenon?	We can explain that organisms can be identified and classified accurately by using a standard system based on shared, observable characteristics and tools such as dichotomous keys, so that anyone anywhere can study and communicate about biodiversity. This replaces our Lesson 1 groupings, which varied from group to group.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.